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GPT, THE COUNTEREXAMPLE MACHINE

COUNT EX MACHINA*

Abstract. This document reports over 15 counterexamples found using GPT Pro over the course of 12 months, for problems in combinatorics, number theory, convexity, analysis, and other areas. The counterexamples concern both (reasonably) well-known problems as well as lesser known instances. The counterexamples are accompanied by the repo *https://github.com/suvrit/count-ex-machina*, and we warmly welcome new contributions toward helping building a rich *"open-source countex library."*

1. Introduction

AI assisted mathematics has literally exploded; as of writing this document, the website [AIMath] lists 438 as of Aug. 9, 2026 (and 506 as of Aug. 25, 2026!) proofs and disproofs of a variety of mathematical questions. Our focus in this manuscript is on counterexamples that refute a conjecture or provide a disproof of some problem, providing both a human friendly proof and a machine-verifiable certificate.

Why counterxamples? The typical asymmetry between proving a claim (universal quantification) versus disproving (existence of one counterexample) makes it particularly attractive to prioritize AI-assisted search. And available AI assistants have been strikingly successful in 2026 at constructing counterexample. Ten noteworthy conjectured claimed to be disproved by AI in 2026 include:

(i) the Jacobian conjecture in 3 dimensions by an explicit Keller map (Alpöge, 2026);
(ii) the Erdős unit-distance conjecture, disproved at OpenAI (2026b); with an explicit exponent by Sawin (2026), and a human-verified digest by Alon et al. (2026);
(iii) the Gaussian completely monotone conjecture (Gu and Sellke, 2026);
(iv) conjectured log-convexity of Fisher information along heat flow (Zou et al., 2026);
(v) Litvak's Gaussian-minima conjecture (Kunisky, 2026);
(vi) the pairwise-independent correlation gap (Ramachandra and Natarajan, 2026);
(vii) near-quadratic Elekes–Rónyai expansion (Liu, 2026);
(viii) the Gaussian moments conjecture (Long, 2026a);
(ix) the xz- and Mathieu conjectures for $SU(2)$ (Long, 2026b); and
(x) conjectured Schur positivity of claw-free chromatic symmetric functions (Matherne and Morales, 2026).

But refutation is not the only aspect the models have been exemplary at: the document (OpenAI, 2026a) includes 10 results obtained by OpenAI's internal Astra model, with several results accompanied by machine-checkable Lean certificates.[1] And the noted collection [AIMath] lists several successful proofs. Though a more mixed set of results from problems designed to challenge AI approaches is in the "First Proof" (Abouzaid et al., 2026) project, where it seems so that AI has not yet managed to make breakthroughs. Continuing the topic of conjectures and their mechanical verification, this new project seems like a wonderful resource (Firsching et al., 2026).

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* Project name used to future proof against possible multiple contributors. The first version is authored by Suvrit Sra (`s.sra@tum.de`; TU Munich, Dept. of Math, School of CIT.).

[1] Allegedly, as of this writing, the claimed disproof of the Connes rigidity conjecture is invalid, because the AI system silently used a group with an infinite center, which violates the conjecture's framing.

Our Philosophy. First, a short word on our *raison d'être*. This work started with the author's effort in collecting some of the AI-assisted disproofs we found over the past 12 months. Importantly, we wanted our exposition of the collected results to be human-readable and human-verifiable, ***while honestly crediting AI for the heavy-lifting***;[2]. In some cases, we share refinements that further enrich the discussion. We supplement the exposition with source code that can be used to machine-verify counterexamples. Though ultimately, it is for the joy of understanding mathematics, of proving and disproving claims, and of sharing progress with others that underscores our motivation; the machine-checkable component is there to support this ideal, beyond mere pedantic necessity.

2. Summary of the counterexamples and resolutions

Some simple criteria for including counterexamples emerged when preparing this manuscript. We hope to use these criteria as a rough yardstick for gauging new contributions to the collection. In particular, we propose the following simple criteria:

Definition 2.1 (**Admission criteria**)**.** (1) The refuted statement appeared publicly before the counterexample was sought (e.g., in a paper or preprint, a problem list, a talk, or in a public post, etc.). The public appearance condition excludes statements invented in order to be refuted.
(2) The original hypotheses and quantifiers are fully preserved. Counterexample to a strengthening or any modification of the original problem are not admitted as refutations.[3]
(3) The refutation is independently verifiable, through exact arithmetic, an interval certificate, an analytic calculation, proof assistant, etc.

Note about problems. Most of the statements studied in the current manuscript involve problems that are likely well-known to specialists but not well-known more broadly. Our collection is open to everybody for contributions, as long as the contributions meet the minimalistic criteria listed above. Moreover, we neither enforce nor advocate any particular verifier, but we do expect every contributor to ensure verification. This manuscript is accompanied by a supporting git repo: *https://github.com/suvrit/count-ex-machina*, which is a living repo and will be the definitive resource on any bugfixes, improvements, additions or otherwise, and we welcome contributions.

Table 1: Summary of counterexamples. *[EC]* denotes *Exact Certificate*; *[IC]* denotes *Interval Certificate*.

	Problem statement	Certificate(s)
1.	Borcea–Brändén Problem 35: existence of Johnson polynomial representation for stable polynomials (Borcea et al., 2009)	Explicit polynomial (deg-5) with immanantal obstruction. *[EC]*; human proof Thm. 3.4.
2.	Borcea–Brändén Problem 38: Stable polynomial Permanent-On-Top (Borcea et al., 2009)	Exact Schur-basis expansion; four violating partitions. *[EC]*; human proof Thm. 3.3.
3.	Borcea–Brändén Problem 36: Schur positivity of symmetric stable polynomials (2009)	Determinantal polynomial in Schur-basis with $[s_{(1^5)}]q = -2972$. *[EC]*; human proof Thm. 3.7.
4.	Borcea–Brändén Problem 37: stable Schur inequality (lower endpoint) (2009)	Schur-*positive* member of the same family with $a_{(1^5)} < f^{(1^5)}a_{(5)}$. *[EC]*; human proof Thm. 3.10.

[2]For instance, it is quite unclear from numerous new (clearly AI-assisted) works appearing on the arXiv whether humans are fairly acknowledging AI's role; our repository is dedicated to collecting results that can be largely attributed to AI, though without trying to be either philosophic or pedantic about "who did what?"

[3]It may well be that a modified statement is the more "natural" one, but given the automation involved, to be pedantically correct, we cannot admit any changes, unless the modified version itself satisfies condition 1.

	Problem statement	Certificate(s)
5.	McSwiggen and Sahi (2026, Theorem 2.1): Claimed log- and Schur-convexity of a ratio of Macdonald polynomials	Failure of Schur (and thus also log-) convexity (rank 2). *[EC]*; human proof Thm. 4.2.
6.	(Coffey and Csordas, 2013, Conj. 2.5) Turán inequalities for Jacobi-Theta derivatives $\Phi^{(n)}$	Rigorous interval evaluation for $n = 9$. *[IC]*; (reduction in Thm. 5.2).
7.	Variance-sensitive Matrix Spencer (Bandeira, 2016, Prob. 4.25)	Diagonal family with discrepancy $\Theta(\sqrt{\log n})$. *[EC]*; human proof Thm. 6.2.
8.	Mixed-norm $\|A^T B\|_{p,q}^2 \le \|A^T A\|_{p,p} \|B^T B\|_{q,q}$ for $1 \le p \le q$ (Sah et al., 2020)	Explicit analytic disproof; *[EC]*; human proof Thm. 7.3.
9.	The same question at $B = A$: $\|Z\|_{s,q}^2 \le \|Z\|_{s,s} \|Z\|_{q,q}$ for completely positive Z	Explicit disproof via a rank-2 21×21 PSD matrix; *[IC]*; no closed-form disproof; 7.5).
10.	"Quantum" coupon-collector: $\sum_{\emptyset \neq S} (-1)^{\|S\|-1} X_S^{-1} \overset{?}{\succ} 0$ (Sra, 2017a).	Disproof via rational 3×3 matrices with $n = 6$; *[EC]*; human proof Thm. 8.4.
11.	O'Donnell (2015) $\|A - D\|_1^2 \overset{?}{\le} C \|\lambda(A) - D\|_1$, where A is PSD and $D = \mathrm{diag}(A)$.	Givens-rotation family with a rational lower-bound; *[EC]*; human proof Thm. 9.2.

Additional Examples (relegated to the Appendix)

12.	Courtade (2017): $V(K+L+B)^{1/d} V(B)^{1/d} + V(K)^{1/d} V(L)^{1/d} \le V(K+B)^{1/d} V(L+B)^{1/d}$.	Disproof in $\mathbb{R}^4$: zonotope plus two segments; *[EC]*; human proof Thm. A.2.
13.	Mariet and Sra (2015)'s feasible-step for a Picard iteration for DPP likelihood ascent.	Exact 2×2 feasible update violating likelihood ascent. *[EC]*; human proof Thm. A.10.
14.	The divergence $\log \frac{V[(X+Y)/2]}{\sqrt{V(X)V(Y)}}$ is a squared distance for convex bodies (Sra, 2013).	Cubic realizable as a volume polynomial; violates triangle inequality. *[EC]*; human proof Thm. A.6.
15.	Lorentzian Jensen–Bregman metrics: $\sqrt{\mathrm{J}_G}$ a semimetric for all homogeneous Lorentzian G.	Disproof via same cubic viewed as a Lorentzian polynomial. *[EC]*; human proof Cor. A.7.
16.	An SDD matrix has a marginal Nyström error reduction(Amsel et al., 2026, Prob. 4.6(b)).	Disproof via an integer 4×4 matrix. *[EC]*; human proof Thm. A.12.

3. Borcea–Brändén AIM problems

3.1. Problems 35 and 38: mixed-determinant representation and Permanent-On-Top.

Posed by Borcea and Brändén (2009); counterexample found by GPT-5 (Pro) (2026-01-10); formalized by S. Sra; audited by S. Sra.

A polynomial $f \in \mathbb{R}[z_1, \dots, z_n]$ is *real stable* if $f(z_1, \dots, z_n) \neq 0$ whenever every z_i has strictly positive imaginary part. A homogeneous symmetric polynomial of degree d expands uniquely in the basis of Schur polynomials as $p = \sum_{\lambda \vdash d} a_\lambda s_\lambda$; we say p is *Schur-positive* if every $a_\lambda \ge 0$. For a partition $\lambda \vdash d$ we write f^λ for the number of standard Young tableaux of shape λ, λ' for the conjugate partition, and, for a $d \times d$ matrix $A = (a_{ij})$,

$$\mathrm{Imm}_\lambda(A) := \sum_{\sigma \in S_d} \chi^\lambda(\sigma) \prod_{i=1}^{d} a_{i\sigma(i)},$$

for the immanant induced by the irreducible character χ^λ of S_d. The two extreme cases are $\mathrm{Imm}_{(1^d)} = \det$ and $\mathrm{Imm}_{(d)} = \mathrm{per}$. Next, for $n \times n$ matrices $A_1, \dots, A_m$ the *mixed-determinant* is

$$\eta(A_1, \dots, A_m) := \sum_{(S_1,\dots,S_m)} \det(A_1[S_1]) \cdots \det(A_m[S_m]),$$

where the sum runs over all ordered partitions of $\{1, \dots, n\}$ into disjoint sets $S_1, \dots, S_m$, and $A[S]$ denotes the principal submatrix of A with rows and columns in S. When $A \succeq 0$ the polynomial $\eta(z_1 A, \dots, z_n A)$ is symmetric, homogeneous, real stable, and has the (Schur-positive) representation

$$\eta(z_1 A, \dots, z_n A) = \sum_{\lambda \vdash n} \mathrm{Imm}_{\lambda'}(A) s_\lambda(z_1, \dots, z_n).$$

The two problems below ask how far those properties go the other way.

Problem 3.1 (As posed: Borcea–Brändén, AIM problem list (2009), Problem 35)**.** Characterize all symmetric, real stable and homogeneous polynomial of degree d in n variables. Is it in fact so that if p is such a polynomial, then there is a $d \times d$ positive semidefinite $d \times d$ matrix A such that

$$p(z_1, \dots, z_n) = \eta(z_1 A, \dots, z_n A)?$$

This is not obvious even for $n = 2$.

Problem 3.2 (As posed: Borcea–Brändén, AIM problem list (2009), Problem 38)**.** Lieb's Permanent-On-Top (POT) Conjecture for the symmetric group on d elements [J,L] asserts that if A is a $d \times d$ positive semi-definite matrix then $\mathrm{Imm}_\lambda(A) \le f^\lambda \,\mathrm{per}(A)$ for all $\lambda \vdash d$. An affirmative answer to the following problem would in particular imply the validity of the POT Conjecture.
Problem 38. Let p be as in Problem 36. Is $a_\lambda \le f^\lambda a_{(1^d)}$ for all $\lambda \vdash d$?

We answers to the above problems are negative. Let $S = x_1 + \cdots + x_5$ and define the polynomial

$$p(x_1, \dots, x_5) = \prod_{i=1}^{5} \Big(x_i + 5 \sum\nolimits_{j \neq i} x_j \Big) = \prod_{i=1}^{5} (5S - 4x_i). \tag{3.1}$$

Every factor has strictly positive coefficients, hence has positive imaginary part whenever all variables do; therefore p is real stable. Moreover, p is symmetric, homogeneous of degree five in five variables, and has strictly positive monomial coefficients.

Theorem 3.3 ([**Disproof**]: Stable Permanent-On-Top, AIM Problem 38)**.** *The polynomial* (3.1) *satisfies all hypotheses of Problem 3.2 but violates* $a_\lambda \le f^\lambda a_{(1^5)}$.

Proof. A short explicit calculation shows that p has the Schur expansion

$$p = 625 s_{(5)} + 4500 s_{(4,1)} + 7125 s_{(3,2)} + 8150 s_{(3,1,1)} + 7525 s_{(2,2,1)} + 6580 s_{(2,1,1,1)} + 1281 s_{(1^5)}.$$

For $\lambda = (3,2)$, the hook-length formula gives $f^{(3,2)} = 5$, whereas

$$7125 > 5 \cdot 1281 = 6405.$$

The same upper endpoint fails for $(3,1,1)$, $(2,2,1)$, and $(2,1,1,1)$. The accompanying source package reconstructs the monomial expansion and inverts the Kostka matrix over $\mathbb{Z}$. □

The polynomial also has a diagonal positive-definite mixed-determinant representation. For $j = 1, \dots, 5$, let D_j denote $5I$ except for the j-th entry being just 1. Then the multivariate Johnson polynomial satisfies

$$\eta(x_1 D_1, \dots, x_5 D_5) = p(x_1, \dots, x_5).$$

Theorem 3.4 ([**Disproof**]: common-matrix representation, AIM Problem 35)**.** *There is no* 5×5 *Hermitian positive semidefinite matrix* A *such that* $p(x_1, \dots, x_5) = \eta(x_1 A, \dots, x_5 A)$.

Proof. Since $\eta(x_1 A, \dots, x_5 A)$ is Schur positive with immanants as coefficients, the representation $p(x) = \eta(x)$ would identify the Schur coefficients (up to normalization). Since Lieb's permanental dominance inequality is known to hold in size five (indeed in size ≤ 13 (Wanless, 2022) it is known), it would force $a_\lambda \leq f^\lambda a_{(1^5)}$ for every $\lambda \vdash 5$. But this contradictions Theorem 3.3, hence there cannot be any PSD A for which $p(x) = \eta(x_1 A, \dots, x_5 A)$. □

Remark 3.5 (Scope). The example is Schur-positive, and its lower endpoint inequalities hold, so it refutes neither Problem 3.6 nor Problem 3.9; those require the positive-definite pencils of Theorems 3.7 and 3.10 below. The full independent audit is included as `counterexamples/aim-problems/artifacts/`.

3.2. **Problem 36: stable Schur positivity.** *Theorem 3.7: Posed by Borcea and Brändén (2009); counterexample found by GPT-5.6 (Pro) (2026-07-30); formalized by S. Sra; audited by S. Sra.*

We keep the notation of the previous section. Problem 3.6 asks whether stability forces a Schur-positive representation.

Problem 3.6 (As posed: Borcea–Brändén, AIM problem list (2009), Problem 36)**.** Let p be a symmetric, real stable and homogeneous polynomial of degree d in n variables with $d \leq n$ and suppose that p has at least one positive (and then all non-negative) coefficients in the monomial basis. Is p Schur-positive? That is, when $p(z_1, \dots, z_n)$ is expanded in the Schur-basis

$$p(z_1, \dots, z_n) = \sum_{\lambda \vdash d} a_\lambda s_\lambda(z_1, \dots, z_n),$$

is $a_\lambda \geq 0$ for all $\lambda \vdash d$?

The answer to Problem 3.6 is no already at $d = n = 5$. Let

$$J_1 = \begin{pmatrix} 38 & 19 & 19 & 2 & 36 \\ 19 & 38 & 2 & 19 & 36 \\ 19 & 2 & 38 & 19 & 36 \\ 2 & 19 & 19 & 38 & 36 \\ 36 & 36 & 36 & 36 & 72 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 38 & 19 & 19 & 17 & 21 \\ 19 & 38 & 17 & 19 & 21 \\ 19 & 17 & 38 & 34 & 36 \\ 17 & 19 & 34 & 38 & 36 \\ 21 & 21 & 36 & 36 & 42 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 38 & 34 & 19 & 17 & 36 \\ 34 & 38 & 17 & 19 & 36 \\ 19 & 17 & 38 & 19 & 21 \\ 17 & 19 & 19 & 38 & 21 \\ 36 & 36 & 21 & 21 & 42 \end{pmatrix},$$

$$J_4 = \begin{pmatrix} 38 & 4 & 19 & 2 & 21 \\ 4 & 8 & 2 & 4 & 6 \\ 19 & 2 & 38 & 4 & 21 \\ 2 & 4 & 4 & 8 & 6 \\ 21 & 6 & 21 & 6 & 42 \end{pmatrix}, \quad J_5 = \begin{pmatrix} 8 & 4 & 4 & 2 & 6 \\ 4 & 38 & 2 & 19 & 21 \\ 4 & 2 & 8 & 4 & 6 \\ 2 & 19 & 4 & 38 & 21 \\ 6 & 21 & 6 & 21 & 42 \end{pmatrix},$$

and define the determinantal polynomial

$$q(x_1, \dots, x_5) = \frac{1}{648} \det\Big(\sum\nolimits_{i=1}^{5} x_i J_i\Big). \tag{3.2}$$

Theorem 3.7 ([**Disproof**]: stable Schur positivity, AIM Problem 36)**.** *The polynomial* (3.2) *is symmetric, homogeneous of degree five in five variables, real stable, and all* 126 *of its ordinary monomial coefficients are strictly positive, yet it is* **not** *Schur positive, since*

$$[s_{(1^5)}]\, q = -2972 < 0.$$

Proof. The leading principal minors of J_i are

	Δ_1	Δ_2	Δ_3	Δ_4	Δ_5
J_1	38	1083	28728	202176	1119744
J_2	38	1083	28728	202176	1119744
J_3	38	288	8208	202176	1119744
J_4	38	288	8208	46656	1119744
J_5	8	288	1728	46656	1119744,

all positive, so Sylvester's criterion gives $J_i \succ 0$; hence q is real stable, and homogeneity of degree 5 is immediate. Exact expansion of the determinant in the monomial basis gives

$$q = 1728m_{(5)} + 19440m_{(4,1)} + 66555m_{(3,2)} + 140910m_{(3,1,1)} + 250440m_{(2,2,1)} + 547405m_{(2,1,1,1)} + 1182860m_{(1^5)}.$$

Inverting the Kostka matrix over $\mathbb{Z}$ in the partition order $(5), (4,1), (3,2), (3,1,1), (2,2,1), (2,1,1,1), (1^5)$ translates q into the Schur basis, yielding

$$q = 1728s_{(5)} + 17712s_{(4,1)} + 47115s_{(3,2)} + 56643s_{(3,1,1)} + 62415s_{(2,2,1)} + 56437s_{(2,1,1,1)} - 2972s_{(1^5)}.$$

The last coefficient alone also follows from the single integer identity

$$[s_{(1^5)}]q = 1728 - 2(19440) - 2(66555) + 3(140910) + 3(250440) - 4(547405) + 1182860 = -2972.$$

The accompanying source package recomputes the Schur expansion in a second, independent way, and every step uses rational arithmetic only. □

Remark 3.8 (Origin of the pencil). The matrices are not a numerical accident. Let $V = \ker B \cong S^{(3,2)}$, where B maps the permutation module on 2-subsets of $[5]$ to $\mathbb{R}^5$ by $e_{ab} \mapsto e_a + e_b$, let P_i be the rank-three projection attached to the point stabilizer of i, and let G be the Gram matrix of the chosen basis of V. Then $J_i = 2G(I + 5P_i)$, and the whole one-parameter family $\det(\sum_i x_i(I + tP_i))$ has a closed Schur expansion whose $s_{(1^5)}$ coefficient turns negative for t large; $t = 5$ clears denominators. *Appendix C develops this construction in full.*

3.3. **Problem 37: Schur-style inequality for stable polynomials.** *Theorem 3.10: Posed by Borcea and Brändén (2009); counterexample found by GPT-5.6 (Pro) (2026-07-30) and bugfixed by Opus 5 (2026-08-10); formalized by S. Sra; audited by S. Sra.*

Next, we close off the lower endpoint analog of Problem 3.2 (which concerned $a_\lambda \le f^\lambda a_{(1^d)}$).

Problem 3.9 (As posed: Borcea–Brändén, AIM problem list (2009), Problem 37)**.** If A is a $d \times d$ matrix then $\det(A) = \mathrm{Imm}_{(1^d)}(A)$ and $\mathrm{per}(A) = \mathrm{Imm}_{(d)}(A)$, where $(1^d) = (1, \dots, 1)$ and $(d) = (1^d)' = (d, 0, \dots, 0)$. Schur's inequality [J,L] asserts that if A is a $d \times d$ positive semi-definite matrix then $\mathrm{Imm}_\lambda(A) \ge f^\lambda \det(A)$ for all $\lambda \vdash d$, where f^λ is the number of standard Young tableaux of shape λ. A natural real stable extension of this result would be as follows.
Problem 37. Let p be as in Problem 36. Is $a_\lambda \ge f^\lambda a_{(d)}$ for all $\lambda \vdash d$?

The lower endpoint needs a different member of the same family, because it needs a witness that is Schur *positive*. Let G be the Gram matrix of the basis of V that produced the J_i,

$$G = \begin{pmatrix} 4 & 2 & 2 & 1 & 3 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 1 & 2 & 2 & 4 & 3 \\ 3 & 3 & 3 & 3 & 6 \end{pmatrix},$$

put $\widetilde{J}_i := (4J_i + 2G)/5$, again a symmetric positive-definite *integer* matrix, and define

$$\widetilde{q}(x_1, \dots, x_5) = \frac{1}{5184} \det\Big(\sum_{i=1}^{5} x_i \widetilde{J}_i\Big), \tag{3.3}$$

which is p_4 in the notation of Appendix C.

Theorem 3.10 ([**Disproof**]: lower endpoint, AIM Problem 37)**.** *The polynomial* (3.3) *satisfies every hypothesis of Problem 3.9, yet it violates the lower endpoint inequality $a_\lambda \geq f^\lambda a_{(d)}$: at $\lambda = (1^5)$, where $f^{(1^5)} = 1$,*

$$a_{(1^5)} = 69 < 125 = a_{(5)}.$$

Proof. The leading principal minors of the $\widetilde{J}_i$ are

	Δ_1	Δ_2	Δ_3	Δ_4	Δ_5
$\widetilde{J}_1$	32	768	17280	118800	648000
$\widetilde{J}_2$	32	768	17280	118800	648000
$\widetilde{J}_3$	32	240	5760	118800	648000
$\widetilde{J}_4$	32	240	5760	32400	648000
$\widetilde{J}_5$	8	240	1440	32400	648000,

all positive, so $\widetilde{J}_i \succ 0$ and $\widetilde{q}$ is real stable, symmetric and homogeneous of degree five by the argument used for (3.2). It can be verified that $\widetilde{q}$ has the Schur positive representation

$$\widetilde{q} = 125s_{(5)} + 1100s_{(4,1)} + 2545s_{(3,2)} + 3110s_{(3,1,1)} + 3321s_{(2,2,1)} + 2972s_{(2,1,1,1)} + 69s_{(1^5)}, \tag{3.4}$$

which can be obtained by starting with the direct monomial basis expansion

$$\widetilde{q} = 125m_{(5)} + 1225m_{(4,1)} + 3770m_{(3,2)} + 7980m_{(3,1,1)} + 13846m_{(2,2,1)} + 30004m_{(2,1,1,1)} + 64472m_{(1^5)},$$

from which upon inverting the Kostka matrix in the same partition order yields (3.4). Now $f^{(1^5)} = 1$, so the lower endpoint at $\lambda = (1^5)$ would require $69 \geq 125$. It is the only partition at which the inequality fails: the remaining six hold, the tightest being $\lambda = (4, 1)$ with $1100 \geq 4 \cdot 125$. □

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4. Schur-convexity of a ratio of Macdonald polynomials

Posed by McSwiggen and Sahi (2026); counterexample found by GPT-5.6 (Pro) (2026-05-14); formalized by S. Sra; audited by S. Sra.

Let $P_\lambda(x; q, t)$ denote the Macdonald polynomial in the monic normalization of Macdonald (1998), with parameters $q, t \in (0, 1)$ over a partition λ with at most n parts. At $q = t$ it specializes to the Schur polynomial $s_\lambda(x)$. The statement concerns two convexity properties of the map $\lambda \mapsto \Omega_\lambda(x; q, t)$ on partitions studied by McSwiggen and Sahi (2026): *log-convexity*, and *Schur-convexity* (i.e., $\lambda \succeq \mu$ implies $\Omega_\lambda \geq \Omega_\mu$). The normalization Ω_λ is defined inside the quoted statement.

Problem 4.1 (As posed: C. McSwiggen and S. Sahi (2026), Theorem 2.1)**.** For all $q, t \in (0, 1)$, $a > 0$, and $x \in \mathcal{L}_n^{q,t,a}$, the function $\lambda \mapsto \Omega_\lambda(x; q, t)$ is log-convex and Schur-convex.
Here, by §2.1 of the same paper, $\Omega_\lambda(x; q, t) = P_\lambda(x; q, t)/P_\lambda(t^\delta; q, t)$ is the Macdonald polynomial normalized to 1 at the principal specialization $t^\delta = (t^{n-1}, t^{n-2}, \dots, 1)$, and the scaled, t-shifted, dominant q-lattice is

$$\mathcal{L}_n = \mathcal{L}_n^{q,t,a} = \Big\{(aq^{-\mu_1}t^{n-1}, aq^{-\mu_2}t^{n-2}, \dots, aq^{-\mu_n}) : (\mu_1 \geq \dots \geq \mu_n) \in \mathbb{Z}^n\Big\} \subset \mathbb{R}^n.$$

Theorem 4.2 ([**Disproof**]: Theorem 2.1 of McSwiggen and Sahi (2026))**.** *The stated Schur-convexity fails already in rank two.*

Proof. Set $q = t = r \in (0,1)$, $a = 1$, and choose lattice index $\nu = (1,0)$. The resulting lattice point is $x = (1,1)$. Let $\lambda = (2,0)$ and $\mu = (1,1)$; then $\lambda \succeq \mu$. At $q = t$, Macdonald polynomials specialize to Schur polynomials, and direct evaluation gives

$$\Omega_{(2,0)}(1,1;r,r) = \frac{3}{1+r+r^2}, \qquad \Omega_{(1,1)}(1,1;r,r) = \frac{1}{r}.$$

But $1+r+r^2-3r = (1-r)^2 > 0$, so $\Omega_{(2,0)} < \Omega_{(1,1)}$ for every $0 < r < 1$, opposite to Schur-convexity. At $r = 1/2$, the values are $12/7 < 2$. □

Remark 4.3 (Scope). The specialization $q = t$ is a convenience. Appendix D carries the reversal for all $q, t \in (0,1)$, with gap $(1-t)^2(1-qt)/(t(1+t)(1-qt^2))$, and shows it is admissible whenever $t = q^k$ for a positive integer k — the case above being $k = 1$. It also locates the failure, in a determinant shift that the principal-specialization normalization does not commute with, and shows that the **1^n-normalized ratio**

$$W_\lambda = P_\lambda(x)/P_\lambda(1^n)$$

satisfies in this case exactly the inequality Ω_λ violates.

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5. Log-concavity of all Derivatives of the Jacobi-theta kernel

Posed by Coffey and Csordas (2014); counterexample found by GPT-5.5 (Pro) (2026-02-22); formalized by S. Sra; audited by S. Sra.

The kernel Φ below is the Jacobi theta kernel whose cosine transform is the Riemann ξ-function. A positive function g is *strictly log-concave* when $(g')^2 - gg'' > 0$, so the assertion $J_n(t) > 0$ says that the $(n-1)$st derivative $\Phi^{(n-1)}$ is strictly log-concave.

Problem 5.1 (As posed: G. Csordas (2014), Open Problem 4.13, restating Coffey–Csordas Conjecture 2.5)**.** ([5, Conjecture 2.5]) Show that the derivatives of the Jacobi theta function, $\Phi(t)$, are (strictly) log-concave on $\mathbb{R}$; that is, for each $n \in \mathbb{N}$,

$$J_n(t) := (\Phi^{(n)}(t))^2 - \Phi^{(n-1)}(t)\Phi^{(n+1)}(t) > 0 \quad \text{for } t \in \mathbb{R}.$$

Here, by (4.2) of the same paper, the Jacobi theta function (without the usual factor 4) is $\Phi(t) := \sum_{n\geq 1} \pi n^2 \left(2\pi n^2 e^{4t} - 3\right) \exp\left(5t - \pi n^2 e^{4t}\right)$.

Theorem 5.2 ([**Disproof**]; computer-assisted proof)**.** *At $n = 9$ and $t = 1/50$, one has $J_9(t) < 0$.*

Proof. Put $u = \pi m^2 e^{4t}$ and define integer polynomials

$$P_0(u) = 2u - 3, \qquad P_{k+1}(u) = 4uP_k'(u) + (5 - 4u)P_k(u).$$

Termwise differentiation gives

$$\Phi^{(k)}(t) = \sum_{m\geq 1} \pi m^2 e^{5t-u} P_k(u).$$

Outward-rounded interval summation, with an explicit geometrically bounded tail, yields

$$\Phi^{(8)}(1/50) \approx 2.95940368409056694 \cdot 10^8,$$
$$\Phi^{(9)}(1/50) \approx 5.80442287530267756 \cdot 10^9,$$
$$\Phi^{(10)}(1/50) \approx 2.02040519425950258 \cdot 10^{11},$$

and an interval for $J_9(1/50)$ contained strictly in the negative half-line, centered at

$$-2.61006208371358922 \cdot 10^{19}.$$

The source package contains a high-precision evaluator and a Sage `RealIntervalField` certificate. The negative margin is many orders of magnitude larger than the certified truncation error. □

Remark 5.3. GPT-5.5 managed to prove the Turán inequalities for $\Phi^{(n)}$ for $n = 2$ and $n = 3$; while we were attempting an inductive proof to generalize those proofs to all n is when the counterexample popped up. However, we have not yet verified whether the Turán inequalities hold for all $n \leq 8$.

6. Variance-sensitive Matrix Spencer

Posed by A. S. Bandeira (2016); counterexample found by GPT-5.5 (Pro) (2026-05-24); formalized by E. Akbaş and S. Sra (2026); audited by S. Sra; contributed by S. Sra.

For symmetric matrices $A_1, \dots, A_n$ the *discrepancy* is $\operatorname{disc}(A_1, \dots, A_n) = \min_{x \in \{\pm 1\}^n} \| \sum_{i=1}^n x_i A_i \|_{\mathrm{op}}$. Spencer's theorem gives an $O(\sqrt{n})$ bound in the scalar case (and thus also for the commuting matrix case); the Matrix Spencer conjecture asks for the same $O(\sqrt{n})$ bound for n symmetric contractions of size $n \times n$. The variance-sensitive strengthening replaces the ambient scale $\sqrt{n}$ by the intrinsic variance parameter $\| \sum_{i=1}^n A_i^2 \|_{\mathrm{op}}^{1/2}$, which never exceeds $\sqrt{n}$ and is often far smaller; it is the parameter that controls the norm of a random signing in matrix concentration, and Kyng et al. (2020) establish the strengthened bound in the rank-one case.

Problem 6.1 (As posed: Remark 4.25 of A. S. Bandeira's problem collection (2016).)**.** There exists a universal constant C, such that for any choice of n symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ satisfying $\|A_i\|_{\mathrm{op}} \leq 1$ there exists a coloring $x \in \{\pm 1\}^n$, such that

$$\left\| \sum_{i=1}^n x_i A_i \right\|_{\mathrm{op}} \leq C \left\| \sum_{i=1}^n A_i^2 \right\|_{\mathrm{op}}^{1/2}.$$

The disproof recorded here is not a new contribution of the present manuscript: the family reported below is taken from Theorem A.1 of Akbaş and Sra (2026), the first refutation of the conjecture; but we couldn't resist including it here as an illustration.

Fix an integer $m \geq 2$ and set $n = 2^m$, so that the matrices below are $n \times n$ and there are n of them: the dimension is not inflated relative to the number of summands. Index coordinates by $s = (s_1, \dots, s_m) \in S := \{\pm 1\}^m$, so $|S| = n$. Write $p = (1, \dots, 1)$ and $q = (-1, \dots, -1)$, choose any $U \subseteq S \setminus \{p, q\}$ with $|U| = m$, put $F = S \setminus U$, and fix a bijection $\pi : \{m+1, \dots, n\} \to F$; note that $p, q \in F$. Define diagonal matrices

$$(A_i)_{s,s} = \frac{s_i}{\sqrt{m}} \quad (1 \leq i \leq m), \qquad A_k = \frac{1}{\sqrt{m}} E_{\pi(k),\pi(k)} \quad (m < k \leq n), \tag{6.1}$$

where $E_{r,r}$ is the diagonal matrix with a single 1 in coordinate r. Each $\|A_i\|_{\mathrm{op}} = 1/\sqrt{m} \leq 1$, so the family is admissible.

Theorem 6.2 (**[Disproof]**: variance-sensitive Matrix Spencer)**.** *For the family (6.1),*

$$\operatorname{disc}(A_1, \dots, A_n) = \frac{m-1}{\sqrt{m}}, \qquad \left\| \sum_{i=1}^n A_i^2 \right\|_{\mathrm{op}}^{1/2} = \sqrt{1 + \tfrac{1}{m}},$$

so the ratio of the two sides equals $(m-1)/\sqrt{m+1} = \Theta(\sqrt{\log n})$*. No universal constant C can therefore satisfy the variance-sensitive bound, already for n matrices of size $n \times n$.*

Proof. Since the matrices are diagonal, so is $\sum_i A_i^2$, and its sth entry is

$$\sum_{i=1}^{m} \frac{s_i^2}{m} + \sum_{k=m+1}^{n} \frac{1}{m}\mathbf{1}\{s = \pi(k)\} = 1 + \frac{1}{m}\mathbf{1}\{s \in F\},$$

whence $\|\sum_i A_i^2\|_{\mathrm{op}} = 1 + \frac{1}{m}$.

For the discrepancy, let $x \in \{\pm 1\}^n$ be an arbitrary signing and $X = \sum_{i=1}^n x_i A_i$, so that

$$X_{s,s} = \frac{1}{\sqrt{m}} \sum_{i=1}^{m} x_i s_i + \frac{1}{\sqrt{m}}\mathbf{1}\{s \in F\} x_{\pi^{-1}(s)}.$$

Evaluating at the coordinate $s = (x_1, \ldots, x_m) \in S$ makes the first sum equal to m, so

$$\|X\|_{\mathrm{op}} \geq |X_{s,s}| \geq \frac{m}{\sqrt{m}} - \frac{1}{\sqrt{m}} = \frac{m-1}{\sqrt{m}}.$$

This bound is attained. Take $x_1 = \cdots = x_m = 1$ and $x_{k_p} = -1$, $x_{k_q} = 1$, where $k_p = \pi^{-1}(p)$ and $k_q = \pi^{-1}(q)$, with the remaining signs arbitrary. Then $\sqrt{m}X_{s,s} = \sum_{j=1}^m s_j + \mathbf{1}\{s \in F\} x_{\pi^{-1}(s)}$, which equals $m-1$ at $s = p$ and $-(m-1)$ at $s = q$. For every other s the vector is neither all-plus nor all-minus, so $|\sum_j s_j| \leq m-2$ and hence $|\sqrt{m}X_{s,s}| \leq m-1$. Thus $\|X\|_{\mathrm{op}} = (m-1)/\sqrt{m}$, and the infimum over signings is exactly that.

Dividing the two displayed quantities gives $(m-1)/\sqrt{m+1}$, which grows without bound as $m = \log_2 n \to \infty$. □

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7. A mixed-norm Cauchy–Schwarz question of Sah, Sawhney, Stoner and Zhao

Posed by A. Sah et al (2020), and put to S. Sra by Y. Zhao in private communication, May 2018; counterexample found by GPT-5.6 Sol (Pro) (2026-07-13); formalized by S. Sra; audited by S. Sra.

Theorem 7.3: Posed by A. Sah et al (2020), and put to S. Sra by Y. Zhao in private communication, May 2018; counterexample found by Opus 5 (2026-08); formalized by S. Sra; audited by S. Sra.

Theorem 7.5: Posed by A. Sah et al (2020), and put to S. Sra by Y. Zhao in private communication, May 2018; counterexample found by GPT-5.6 Sol (Pro) (2026-07-13); formalized by S. Sra; audited by S. Sra.

For a matrix $X \in \mathbb{C}^{m\times n}$ the $\ell_{p,q}$ *mixed norm* is defined as:

$$\|X\|_{p,q} = \left(\sum_{j=1}^{n} \left(\sum_{i=1}^{m} |x_{ij}|^p\right)^{q/p}\right)^{1/q}.$$

A matrix is *completely positive* if it has the factorization YY^T for some entrywise nonnegative Y.

Problem 7.1 (As posed: A. Sah, M. Sawhney, D. Stoner and Y. Zhao, the open question stated immediately after inequality (3.1) in §3)**.** We do not know if the inequality can be extended to $\|A^{\mathsf{T}}B\|_{p,q}^2 \leq \|A^{\mathsf{T}}A\|_{q,q}\|B^{\mathsf{T}}B\|_{p,p}$ for all reals $1 \leq p \leq q$ (this is true for positive integer p by a tensor-power argument).

Problem 7.2 (As posed, paraphrased: A. Sah, M. Sawhney, D. Stoner and Y. Zhao, the same open question at $B = A$; shared with S. Sra by Y. Zhao in private communication, May 2018)**.** Taking $B = A$ in the question above and writing $Z = AA^{\mathsf{T}}$ for the resulting completely positive matrix: for all reals $1 \leq s \leq q$ and every completely positive Z,

$$\|Z\|_{s,q}^2 \leq \|Z\|_{s,s}\|Z\|_{q,q}.$$

Both Problems 7.1 and 7.2 fail by witnesses of quite different character. Problem 7.1 fails at every non-integer exponent as a consequence of the Hadamard-power theory of FitzGerald and Horn (1977) and Horn and Mathias (1990, Theorem 4.3); that argument is recorded as Remark 7.4 below (and it is my own observation, not a connection due to an AI assistant). Theorem 7.3 adds a minimal explicit witness though. The second witness is larger and interval-certified; it refutes the harder $B = A$ specialization of Problem 7.2. We note that for $p = 1$ (and hence every integer valued p by tensoring) the alleged inequalities do hold—*see Appendix B for details and additional results.*

Theorem 7.3 ([**Disproof**]: explicit disproof, $A \neq B$)**.** *Let $p = q = 3/2$ and put*

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 0 \\ 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 5 \\ 4 & 2 \\ 1 & 0 \end{bmatrix} \in \mathbb{Z}_{\geq 0}^{3\times 2}.$$

Then $\|A^{\mathsf{T}}B\|_{p,q}^2 > \|A^{\mathsf{T}}A\|_{q,q}\|B^{\mathsf{T}}B\|_{p,p}$.

Proof. Both matrices are entrywise nonnegative, so they satisfy the hypothesis of the question as posed, not merely the weaker variant in which only the three products are required to be nonnegative. Those products are

$$A^{\mathsf{T}}A = \begin{bmatrix} 30 & 8 \\ 8 & 16 \end{bmatrix}, \qquad B^{\mathsf{T}}B = \begin{bmatrix} 17 & 8 \\ 8 & 29 \end{bmatrix}, \qquad A^{\mathsf{T}}B = \begin{bmatrix} 21 & 20 \\ 0 & 20 \end{bmatrix}.$$

At $p = q = 3/2$ we calculate $\|M\|_{3/2,3/2}^{3/2} = \sum_{ij} |m_{ij}|^{3/2}$, and the deficit

$$\Delta := \|A^{\mathsf{T}}A\|_{3/2,3/2}^{3/2} + \|B^{\mathsf{T}}B\|_{3/2,3/2}^{3/2} - 2\|A^{\mathsf{T}}B\|_{3/2,3/2}^{3/2}$$

is an integer combination of square roots of integers:

$$\Delta = 64 + 64\sqrt{2} + 30\sqrt{30} + 17\sqrt{17} + 29\sqrt{29} - 42\sqrt{21} - 160\sqrt{5}.$$

Deciding the sign of Δ is exact algebra over a real field, with no rounding anywhere:

$$\Delta = -5.150045302282868288532184137690370433502833719528\ldots < 0.$$

Hence $2\|A^{\mathsf{T}}B\|_{3/2,3/2}^{3/2} > \|A^{\mathsf{T}}A\|_{3/2,3/2}^{3/2} + \|B^{\mathsf{T}}B\|_{3/2,3/2}^{3/2}$, and the AM-GM inequality bounds the right side below by $2\big(\|A^{\mathsf{T}}A\|_{3/2,3/2}\|B^{\mathsf{T}}B\|_{3/2,3/2}\big)^{3/4}$. Canceling and raising to the power $2/3$,

$$\frac{\|A^{\mathsf{T}}B\|_{3/2,3/2}}{\|A^{\mathsf{T}}A\|_{3/2,3/2}^{1/2}\|B^{\mathsf{T}}B\|_{3/2,3/2}^{1/2}} = 1.00629360763001504125\ldots > 1. \qquad \square$$

Remark 7.4 (Why non-integer exponents fail). Let $s \notin \mathbb{N}$ with $s < n - 2$. By the sharpness half of the FitzGerald–Horn theorem (FitzGerald and Horn, 1977) there is a positive semidefinite $Z \in \mathbb{R}_{\geq 0}^{n\times n}$ whose Hadamard power $Z^{\odot s}$ is not positive semidefinite, so $\sigma^{\mathsf{T}}Z^{\odot s}\sigma < 0$ for some σ. Rescaling if necessary, we can take σ to be in $\{\pm 1\}^n$; then, splitting the index set along the sign of σ write Z as the block Gram matrix of two nonnegative blocks A, B, and the displayed quadratic form is exactly $-\Delta$. This is the mechanism of Horn and Mathias (1990, Theorem 4.3), and it produces a counterexample for *every* non-integer s.

Theorem 7.5 ([**Disproof**]; computer-assisted disproof $A = B$ case)**.** *There is an explicit 21×21 completely positive matrix Z of rank two with*

$$\|Z\|_{s,q}^2 > \|Z\|_{s,s}\|Z\|_{q,q}, \qquad (s, q) = (6/5, 6).$$

Proof. Let X have 21 rows, grouped as follows. For each group use the indicated multiplicity m_g, weight w_g, and direction y_g:

g	1	2	3	4
m_g	1	18	1	1
w_g	46	17	42	1
y_g	$(1,0)$	$(1,1/50)$	$(1,3/50)$	$(0,1)$.

Every row in group g is $w_g^{5/6} y_g$, and set $Z = XX^{\mathsf{T}}$. Thus $Z \succeq 0$, Z is entrywise nonnegative, and $\operatorname{rank} Z = 2$; being a Gram matrix of nonnegative vectors, Z is completely positive. Put

$$R_Z = \frac{\|Z\|_{6/5,6}}{\sqrt{\|Z\|_{6/5,6/5}\|Z\|_{6,6}}}.$$

Direct interval evaluation gives

$$R_Z \geq 1.00000061733911154365777590600087863798,$$
$$R_Z \leq 1.00000061733911154365777590600087863800.$$

The entire interval lies above one. The source package regenerates Z from the four groups and evaluates the ratio independently at 100 digits; a Sage interval script supplies outward rounding. □

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8. Quantum coupon collection: an alternating sum of inverses

Posed by S. Sra (2017a); counterexample found by GPT Pro (2026-02-16); formalized by GPT-5.6 Pro, which re-derived the exact witnesses in 2026-08; audited by GPT-5.6 Pro; contributed by S. Sra.

In the classical coupon collector's problem with sampling probabilities $x_1, \ldots, x_n > 0$, the expected waiting time is the alternating sum

$$T_n(x_1, \ldots, x_n) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} \frac{1}{x_{i_1} + \cdots + x_{i_k}},$$

whose positivity can be seen through a clever trick using the fact that $t \mapsto 1/t$ is completely monotone on $(0, \infty)$ (Flajolet et al., 1992). The statement quoted below asks whether the same holds with the positive numbers replaced by positive definite matrices.

Problem 8.1 (As posed: S. Sra, MathOverflow question 263833 (2017a))**.** Here we take symmetric positive definite matrices $X_1, \ldots, X_n \in \mathbf{S}^d_{++}$, and consider the sum

$$Q_n(X_1, \ldots, X_n) := \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} (X_{i_1} + \cdots + X_{i_k})^{-1}.$$

Conjecture. For $n \geq 1$, and $X_1, \ldots, X_n \in \mathbf{S}^d_{++}$, we have $Q_n(X_1, \ldots, X_n) \succ 0$.
Note. The cases $n = 1, 2, 3$ are trivially true. I've tried $n = 4$ numerically, and it seems to hold. (Also, by CCP, positivity immediately holds for simultaneously diagonalizable matrices.)

The conjecture holds for a tuple of simultaneously diagonalizable matrices, where it is the classical identity coordinate by coordinate, and the source records that $n \leq 3$ is trivial and that $n = 4$ was checked numerically. Lemma 8.2 and Theorem 8.3 below upgrade that record: the conjecture is a theorem for every d and every $n \leq 5$. It fails at $n = 6$. Let us first present proofs of when it holds before a disproof of the general case.

Lemma 8.2 (Facet averaging)**.** *Let $A_1, \dots, A_m \in \mathbf{S}^d_{++}$, where $m \geq 2$, and put $A = \sum_{j=1}^m A_j$. Then*

$$A^{-1} \preceq \frac{m-1}{m^2} \sum_{j=1}^m (A - A_j)^{-1}.$$

Proof. Set $B_j = A - A_j$, which is positive definite because it is a sum of the remaining A_i. Since $\frac{1}{m}\sum_{j=1}^m B_j = \frac{m-1}{m} A$ and the inversion map is operator convex on $\mathbf{S}^d_{++}$,

$$\frac{m}{m-1} A^{-1} = \Big(\frac{1}{m}\sum_{j=1}^m B_j\Big)^{-1} \preceq \frac{1}{m}\sum_{j=1}^m B_j^{-1}.$$

Rearranging gives the claim. □

For a nonempty $S \subseteq [n]$ we abbreviate $X_S := \sum_{i \in S} X_i$; then, for fixed n, introduce the layer sums

$$E_k := \sum_{S \subseteq [n], |S| = k} X_S^{-1}, \qquad 1 \leq k \leq n,$$

each of which is positive definite. Applying Lemma 8.2 inside each $(k+1)$-subset T, with the A_j taken to be the X_i for $i \in T$, gives $X_T^{-1} \preceq \frac{k}{(k+1)^2} \sum_{i \in T} X_{T \setminus \{i\}}^{-1}$. Summing over all T of size $k+1$, and noting that each fixed k-subset arises as a facet of exactly $n-k$ of them, yields

$$E_{k+1} \preceq \frac{k(n-k)}{(k+1)^2} E_k, \qquad 1 \leq k < n. \tag{8.1}$$

Theorem 8.3 (**[Proof]**: positivity through five variables)**.** *For every $d \geq 1$, every $1 \leq n \leq 5$, and every $X_1, \dots, X_n \in \mathbf{S}^d_{++}$, one has $Q_n(X_1, \dots, X_n) \succ 0$.*

Proof. $n = 1$ is immediate. Write $Q_n = E_1 - E_2 + E_3 - \cdots + (-1)^{n-1} E_n$ and apply (8.1) to obtain:

$$\begin{aligned}
n = 2: &\quad E_2 \preceq \tfrac{1}{4} E_1, && Q_2 \succeq \tfrac{3}{4} E_1, \\
n = 3: &\quad E_2 \preceq \tfrac{1}{2} E_1, && Q_3 \succeq \tfrac{1}{2} E_1 + E_3, \\
n = 4: &\quad E_2 \preceq \tfrac{3}{4} E_1, \quad E_4 \preceq \tfrac{3}{16} E_3, && Q_4 \succeq \tfrac{1}{4} E_1 + \tfrac{13}{16} E_3, \\
n = 5: &\quad E_2 \preceq E_1, \quad E_4 \preceq \tfrac{3}{8} E_3, && Q_5 \succeq \tfrac{5}{8} E_3 + E_5.
\end{aligned}$$

Each displayed lower bound is clearly positive definite. □

Theorem 8.4 (**[Disproof]**: quantum coupon collection)**.** *The conjecture is false: there are $d = 3$, $n = 6$ and $X_1, \dots, X_6 \in \mathbf{S}^3_{++}$ with $Q_6(X_1, \dots, X_6) \not\succeq 0$, so in particular $Q_6 \succ 0$ fails.*

Proof. Take $d = 3$, $n = 6$, $\varepsilon = 1/100$, and

$$X_i = w_i u_i u_i^{\mathsf{T}} + \varepsilon I_3,$$

where

i	u_i^{T}	w_i
1	$(-1, 2, 1)$	10
2	$(0, -3, 1)$	10
3	$(-3, -2, 0)$	100
4	$(2, -2, -2)$	100
5	$(-1, 0, 2)$	100
6	$(-1, 3, 0)$	100

Each X_i is strictly positive definite because $X_i \succeq \varepsilon I_3 \succ 0$, so the hypotheses are met exactly as posed. The example is genuinely noncommutative — the $(1,2)$ entry of $[X_1, X_2]$ equals -1500 — and therefore lies outside the simultaneously diagonalizable regime in which the classical identity applies. For $v = (4, 1, 3)^{\mathsf{T}}$, exact rational evaluation of the sixty-three nonempty-subset terms gives

$$-96 < v^{\mathsf{T}} Q_6(X_1, \dots, X_6) v < -95.$$

The accompanying verifier performs this computation using only rational arithmetic. Thus, Q_6 has a negative quadratic form and is not positive semidefinite, let alone positive definite. □

The failure is not confined to the matrix ordering: the weaker scalar consequence $\operatorname{tr} Q_n > 0$ fails as well; *see Appendix E for more details and commentary.*

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9. Ryan O'Donnell's matrix conjecture

Posed by R. O'Donnell and J. Wright (2015); counterexample found by GPT-5.6 Pro (2026-08); formalized by GPT-5.6 Pro; audited by GPT-5.6 Pro; contributed by S. Sra.

For a Hermitian matrix X write $\|X\|_1 = \operatorname{tr}|X|$ for its trace norm. For $R \in \mathbb{C}^{n\times n}$, let $D = \operatorname{Diag}(R) = \operatorname{diag}(d_1, \dots, d_n)$ be its diagonal part, and let $\Lambda(R)$ be the diagonal matrix carrying the eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$ in sorted order, so that

$$\epsilon := \|D - \Lambda(R)\|_1 = \sum\nolimits_i |\lambda_i - d_{ii}|.$$

The source states its conclusion asymptotically, as $\|R - D\|_1 = O(\sqrt{\epsilon})$; an absolute constant in that $O(\cdot)$ is exactly an absolute c with $\|R - D\|_1^2 \leq c\,\epsilon$, which is the form refuted below.

Problem 9.1 (As posed, paraphrased: R. O'Donnell, answering a question of Nengkun Yu on MathOverflow (O'Donnell, 2015), reports that he and J. Wright had been considering the problem in connection with quantum tomography, and closes by conjecturing the unit-trace case. The same conjecture was put to S. Sra by O'Donnell in personal communication.)**.** The question answered was whether a positive semidefinite M with $\|D(M) - \Lambda(M)\|_1 \leq \epsilon$ must lie within 2ϵ of conjectures the normalized case:

> "If you restrict M to have trace 1, then we felt the desired result was true but with a weaker conclusion of $O(\sqrt{\epsilon})$ rather than 2ϵ."

Written out, and with the implied constant absolute: for every unit-trace positive semidefinite R with eigenvalues λ_i and diagonal entries d_{ii},

$$\|R - \operatorname{Diag}(R)\|_1^2 \leq c \sum\nolimits_i |\lambda_i - d_{ii}|$$

for some absolute constant c independent of n.

Theorem 9.2 ([**Disproof**]: O'Donnell's matrix conjecture)**.** *For every $C > 0$ there exist an integer n and a PSD matrix $R \in \mathbb{R}^{n\times n}$ with $\operatorname{tr} R = 1$ such that, writing $D = \operatorname{Diag}(R) = \operatorname{diag}(d_1, \dots, d_n)$ and listing the eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$,*

$$\frac{\|R - D\|_1^2}{\sum_{i=1}^n |\lambda_i - d_i|} > C.$$

Moreover, both (λ_i) and (d_i) are strictly decreasing.

Proof. Fix an integer $m \geq 2$ and put $n = 2^m$, $q = n$, and $H = H_{n-1} = \sum_{i=1}^{n-1} \frac{1}{i}$. Then, set $B = 1 + q^2 H$, $\delta = \frac{1}{2B}$, and $\eta = \frac{1}{2n}$, and define positive spectral gaps $g_1 = \delta(1 + q^2)$, $g_i = \delta q^2/i^2$ for $(2 \leq i \leq n-1)$. Then set the eigenvalues as

$$\lambda_n = \eta, \qquad \lambda_i = \eta + \sum_{k=i}^{n-1} g_k \quad (1 \leq i \leq n-1).$$

Clearly, $\lambda_1 > \cdots > \lambda_n > 0$, and $\sum_{i=1}^n \lambda_i = n\eta + \sum_{i=1}^{n-1} i g_i = \frac{1}{2} + \delta\left(1 + q^2 + q^2 \sum_{i=2}^{n-1} \frac{1}{i}\right) = \frac{1}{2} + \delta B = 1$.

Now for $1 \le i \le n-1$, put

$$r_i = \frac{q}{i}, \qquad c_i = \frac{q}{\sqrt{q^2+i^2}}, \qquad s_i = \frac{i}{\sqrt{q^2+i^2}}.$$

Hence $c_i^2 + s_i^2 = 1$ and $c_i/s_i = r_i$. Let G_i be the identity except for the following orthogonal block in coordinates $i, i+1$:

$$\begin{pmatrix} c_i & -s_i \\ s_i & c_i \end{pmatrix}.$$

Starting from $R^{(0)} = \Lambda$, define successively

$$R^{(i)} = G_i R^{(i-1)} G_i^{\mathsf{T}}, \qquad 1 \le i \le n-1,$$

and let $R = R^{(n-1)}$. This is an exact algebraic matrix. Since it is orthogonally similar to Λ, it is real symmetric, positive semidefinite, has trace one, and has eigenvalues (λ_i).

We next compute its diagonal exactly. Before the first rotation, the active diagonal gap is

$$\lambda_1 - \lambda_2 = g_1 = \delta(1 + r_1^2).$$

Before rotation G_i for $i \ge 2$, coordinate i has diagonal entry $\lambda_i + \delta$, coordinate $i+1$ has diagonal entry λ_{i+1}, and coordinate $i+1$ has not previously been touched. Thus the active 2×2 principal block is diagonal and its diagonal gap is

$$(\lambda_i + \delta) - \lambda_{i+1} = g_i + \delta = \delta(1 + r_i^2).$$

In both cases, $s_i^2\, \delta(1 + r_i^2) = \delta$. Consequently, G_i transfers exactly δ from diagonal position i to position $i+1$. Induction gives

$$d_1 = \lambda_1 - \delta, \qquad d_i = \lambda_i \quad (2 \le i \le n-1), \qquad d_n = \lambda_n + \delta.$$

These entries are strictly decreasing: at the first edge the remaining gap is $g_1 - \delta = \delta q^2 > 0$, at every interior edge it is $g_i > 0$, and at the last edge it is

$$g_{n-1} - \delta = \delta \left(\frac{q^2}{(n-1)^2} - 1 \right) > 0$$

because $q = n$. Therefore $\sum_{i=1}^{n} |\lambda_i - d_i| = 2\delta$. The same rotation calculation gives useful off-diagonal entries. At step i, the newly created $(i, i+1)$ entry has magnitude $c_i s_i\, \delta(1 + r_i^2) = \delta r_i$. For $i \le n-2$, the next rotation multiplies this entry by c_{i+1}; all later rotations leave it unchanged. Hence

$$|R_{i,i+1}| = \begin{cases} \delta r_i c_{i+1}, & 1 \le i \le n-2, \\ \delta r_{n-1}, & i = n-1. \end{cases}$$

Let $A = R - D$. Pinch A to the disjoint coordinate blocks $(1,2), (3,4), \ldots, (n-1, n)$. Trace norm is contractive under pinching, and each retained zero-diagonal 2×2 block has trace norm twice the magnitude of its off-diagonal entry. If

$$O = \sum_{\substack{1 \le i \le n-1 \\ i \text{ odd}}} \frac{1}{i},$$

then $q = n$ implies $c_j^2 = \frac{q^2}{q^2 + j^2} > \frac{1}{2} \qquad (1 \le j \le n-1)$. Using the multiplier $1 > 1/\sqrt{2}$ for the final edge as well, we obtain

$$\|A\|_1 \ge 2 \sum_{\substack{1 \le i \le n-1 \\ i \text{ odd}}} |R_{i,i+1}| > \sqrt{2}\, \delta q O.$$

It follows that

$$\frac{\|R - D\|_1^2}{\sum_i |\lambda_i - d_i|} > \frac{2\delta^2 q^2 O^2}{2\delta} = \delta q^2 O^2 = \frac{q^2 O^2}{2(1 + q^2 H)} \quad > \quad \frac{O^2}{2(H+1)}.$$

It remains only to estimate the two harmonic sums. In each dyadic block $[2^k, 2^{k+1})$, for $1 \le k \le m-1$, there are 2^{k-1} odd integers, each contributing more than $2^{-(k+1)}$. Therefore, $O > 1 + \frac{m-1}{4} = \frac{m+3}{4}$, and also $H_{n-1} < 1 + \log(n-1) < m+1$. Consequently,

$$\frac{\|R-D\|_1^2}{\sum_i |\lambda_i - d_i|} > \frac{(m+3)^2}{32(m+2)} > \frac{m}{32}.$$

Given any real $C > 0$, choose an integer $m > 32C$. The resulting exact matrix has ratio greater than C, proving that no dimension-independent constant can exist. □

10. Conclusion

The archive behind this manuscript currently holds eighteen refuted statements across fourteen cases; sixteen of them reduce to exact rational arithmetic and two to rigorous interval enclosures, and every one recomputes from source in the accompanying repository (several of the refutations are in additional supported by an analytic proof too, not just numerical).

What we find most interesting, though, is that the better counterexamples say more than "false." They locate a boundary. The quantum coupon-collector sum is positive definite through five variables in every dimension and fails at six; the mixed-norm inequality holds at every positive integer exponent and fails at every non-integer one; and in the same rank-two instance where Schur convexity of the t^δ-normalized Macdonald ratio fails, the 1^n-normalized ratio satisfies the very inequality violated. A refutation of that kind does not close a subject so much as move it, and the corrected question it leaves behind is usually the one worth asking next.

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Statement on AI Usage. I used Claude Code (Opus 5) to architect the document and the repo. Initial drafts were assembled by Claude using some of my GPT Pro transcripts as well as from my own existing LaTeX code. However, I rewrote the bulk of the main document (excluding the appendix) myself to improve clarity and precision, hoping to also retain *"my own voice."* The bulk of the credit for the counterexamples reported herein goes to GPT. Moreover, to avoid meta-discourse I'm refraining from dwelling too much on the session prompts—some counterexamples were found in a response to a single-shot prompt, others after an elaborate chat session. That said, I've taken the opportunity to piggyback onto this document some related results that are interesting enough to share and were actually found "by hand"—I was too lazy to write up as separate preprints, so now they are reported in the appendix; their descriptive text notes their human origin.

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Appendix A. Additional counterexamples

We list some additional counterexamples below. They are not in the main body, either because by the time we wrote them down others had also found their own counterexamples, or the statement under study did not meet all our admission criteria (e.g., public availability of the conjecture).

A.1. Courtade's volume conjecture for Minkowski sums is false.

Posed by Thomas A. Courtade (2017); counterexample found by Suvrit Sra, by hand (2022-05); formalized by Claude Fable 5; audited by Claude Fable 5; contributed by Suvrit Sra.

For two sets $K, L \subset \mathbb{R}^d$ the Minkowski sum is the set $K + L := \{x + y : x \in K,\ y \in L\}$. The statement below is the closing question of Courtade (2017); the "Theorem 1" refers to Courtade's similarly structured entropy-power inequality, of which he conjectured a volume analogue. Dropping $V(K)^{1/d}V(L)^{1/d}$ weakens his question to the *log-submodularity* of volume under Minkowski summation, $V(B)V(K+L+B) \le V(K+B)V(L+B)$, a property with its own literature; see the remarks after the proof. Following Courtade, we'll locally also use the notation $|K| \equiv V(K)$.

Problem A.1 (As posed: Closing remarks (Section IV) of Courtade (2017), posed as a question for future work; no problem number is assigned there.)**.** In particular, for K, L, B bounded convex sets in $\mathbb{R}^d$, does it hold that:

$$|K+L+B|^{1/d}|B|^{1/d} + |K|^{1/d}|L|^{1/d} \le |K+B|^{1/d}|L+B|^{1/d},$$

where $|\cdot|$ denotes d-dimensional volume?

Problem A.1 asks for the geometric analogue of the entropy-power inequality that Courtade (2017) proves for Gaussian perturbations: volumes of Minkowski sums should play the role of entropy powers of independent sums. For $d = 1$ the two sides agree identically, and for $d = 2$ the inequality is true (Fradelizi et al., 2024a). It fails from dimension three onwards. We give a fully explicit witness in the zonotope calculus, where every volume in sight is an integer.

Theorem A.2 ([**Disproof**]: Courtade's volume conjecture)**.** *For every $d \geq 4$ there are bounded convex sets $K, L, B \subset \mathbb{R}^d$ with*

$$|K + L + B|^{1/d}|B|^{1/d} + |K|^{1/d}|L|^{1/d} > |K + B|^{1/d}|L + B|^{1/d}.$$

One may take B a zonotope and K, L segments; in $\mathbb{R}^4$, B has seven integer generators and all three sets may moreover be taken full-dimensional.

Proof. A zonotope $Z(v_1, \ldots, v_m) = \sum_{i \leq m}[0, v_i] \subset \mathbb{R}^d$ has volume $\sum_S |\det(v_S)|$, the sum running over all d-element subsets S of the generators (Shephard, 1974), and Minkowski summation simply concatenates generator lists. In $\mathbb{R}^4$ let B be the zonotope with the seven columns of

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

as generators, and let $K = [0, b]$ and $L = [0, c]$ be the segments with $b = (1, -1, 1, 1)$ and $c = (0, 0, -1, 1)$. Summing $|\det|$ over the $\binom{7}{4} = 35$ generator quadruples of B (eleven are singular, twenty have $|\det| = 1$, four have $|\det| = 2$) gives $|B| = 28$, and likewise over the $\binom{8}{4} = 70$ quadruples of $B + K$ and of $B + L$ and the $\binom{9}{4} = 126$ of $B + K + L$:

$$|B| = 28, \qquad |K + B| = 68, \qquad |L + B| = 44, \qquad |K + L + B| = 108.$$

Since $|K| = |L| = 0$, the conjectured inequality asserts $(28 \cdot 108)^{1/4} \leq (68 \cdot 44)^{1/4}$. Raising to the fourth power, which is monotone on nonnegative reals, this is $3024 \leq 2992$ — false.

For full-dimensional sets in $\mathbb{R}^4$, thicken each segment to a box: with $\varepsilon = \frac{1}{100}$ replace K by $K_\varepsilon = [0, b] + [0, \varepsilon]^4$ and L by $L_\varepsilon = [0, c] + [0, \varepsilon]^4$, both zonotopes with five generators and positive volume ($|K_\varepsilon| = \frac{401}{10^8}$, $|L_\varepsilon| = \frac{201}{10^8}$). The same generator sums give the exact rationals

$$|K_\varepsilon + B| = \tfrac{6926731601}{10^8}, \qquad |L_\varepsilon + B| = \tfrac{4484551401}{10^8}, \qquad |K_\varepsilon + L_\varepsilon + B| = \tfrac{696978401}{6250000},$$

whence $|B|\,|K_\varepsilon + L_\varepsilon + B| - |K_\varepsilon + B|\,|L_\varepsilon + B| = \frac{161348459184476999}{10^{16}} > 0$. In fourth powers the conjectured inequality fails a fortiori, its left side now carrying the additional positive term $|K_\varepsilon|^{1/4}|L_\varepsilon|^{1/4}$.

For $d > 4$ pad only the common summand: replace B by $B \times [0, 1]^{d-4}$ and embed K, L as the segments $[0, (b, 0)]$, $[0, (c, 0)]$. The three sums then acquire the same cube factor, so all four volumes above are unchanged, and $|K| = |L| = 0$ still; the same four integers refute the inequality in $\mathbb{R}^d$. Every quantity displayed in this proof is an integer or an exact rational, recomputed by `verify.py` in integer respectively rational arithmetic, and cross-checked in SageMath, which rebuilds each Minkowski sum as a vertex polytope over $\mathbb{Q}$ and computes its volume by triangulation instead. □

The author's own (non AI) proof. Unlike most of this archive, this refutation was first derived by hand, in May 2022 (independently of, and essentially contemporaneously with, the arXiv posting of Fradelizi et al. (2024b) in June 2022; no priority is claimed) by a three-step reduction that shows failure for every $d \geq 3$ without exhibiting a witness. It is the shortest disproof of Problem A.1 we know of, so we record it in full. Throughout, a, b, c, h are convex bodies in $\mathbb{R}^d$, $V(a) = |a|$, and $V(a[d-k], h[k])$ is the mixed volume with a repeated $d - k$ times.

Step 1: drop a term. Applied to $(K, L, B) = (b, c, a)$ and stripped of the nonnegative term $|b|^{1/d}|c|^{1/d}$, the conjecture asserts $V(a)\,V(a+b+c) \le V(a+b)\,V(a+c)$, which rearranges to

$$f_b(a+c) \;\ge\; f_b(a) \qquad \text{for} \qquad f_b :\; a \;\longmapsto\; \frac{V(a)}{V(a+b)} :$$

for every fixed b, the functional f_b must be non-decreasing along Minkowski addition.

Step 2: linearize. The expansion $V(a+th) = \sum_{k\le d} \binom{d}{k}\, t^k\, V(a[d{-}k], h[k])$ is a polynomial in $t \ge 0$ with $\frac{d}{dt} V(a+th)\big|_{t=0} = d\,V(a[d{-}1], h)$, so $t \mapsto \log f_b(a+th)$ is differentiable; by Step 1 (applied at each base point $a+th$, with increment a multiple of h) it is non-decreasing, so its derivative at $t = 0$ is nonnegative:

$$\frac{V(a[d{-}1], h)}{V(a)} \;\ge\; \frac{V((a{+}b)[d{-}1], h)}{V(a+b)} \qquad \text{for all bodies } a, b, h.$$

Step 3: take h a ball. Since $d\,V(a[d{-}1], B_2^d) = |\partial a|$ is surface area, the display at $h = B_2^d$ is exactly the monotonicity $I(a+b) \ge I(a)$ of the information functional $I(\cdot) = |\cdot|/|\partial\,\cdot|$ of Dembo et al. (1991). Artstein-Avidan et al. (2014, Section 4.1) exhibit, for every $d \ge 3$, an interval b and a truncated box a with $I(a+b) < I(a)$ — and prove that monotonicity does hold for $d = 2$, consistent with the planar case of the conjecture being true. Undoing Step 2, the strictly negative derivative gives a $t > 0$ with $V(a+b)\,V(a+tB_2^d) < V(a)\,V(a+b+tB_2^d)$, and the conjecture fails at $(K, L, B) = (b,\, tB_2^d,\, a)$.

The explicit zonotope witness above was constructed for this archive in August 2026, so that the case is self-contained and exactly recomputable; it lands in $d = 4$ of necessity, since log-submodularity of volume holds for zonoids in $d \le 3$ (Fradelizi et al., 2024a).

What holds instead: hyperbolic polynomials Replace $\mathrm{Vol}^{1/d}$ on convex bodies by $p^{1/m}$ on the closed hyperbolicity cone of a hyperbolic polynomial p of degree m. Then, the four-point inequality becomes a theorem, since Theorem F.4 in Appendix F proves

$$p^{1/m}(a+b)\,p^{1/m}(a+c) \;\ge\; p^{1/m}(a)\,p^{1/m}(a+b+c) + p^{1/m}(b)\,p^{1/m}(c),$$

extra term included. The weaker inequality without the $p^{1/m}(b)\,p^{1/m}(c)$ was first observed by the author (S. Sra) here (Sra, 2017b).

Scope, and others' counterexamples. The conjecture has by now been refuted several times over. The sharp Plünnecke–Ruzsa constants of Fradelizi et al. (2024b) already imply that log-submodularity — hence the stronger inequality here — fails for general convex bodies in every dimension $d \ge 3$. In August 2026, Skorupinski (2026) independently gave a zonotope counterexample to log-submodularity in $\mathbb{R}^4$, with volume vector $(|B|, |K{+}B|, |L{+}B|, |K{+}L{+}B|) = (14, 30, 26, 56)$. The witness above came from an independent random search over integer generators, which turned up dozens of violating configurations — among them, rediscovered up to a unimodular transformation, Skorupinski's own witness with its volume vector $(14, 30, 26, 56)$ — so such violations are not isolated. The configuration kept here is a different one: its volume vector $(28, 68, 44, 108)$ is not proportional to his, so the two are not affinely equivalent, and its relative margin is about twice as large. What none of these witnesses touch is the variant of the question in which the common summand B is a Euclidean ball — the case closest to W Gaussian in the entropy-power inequality it mirrors, and one where the linearization above is provably monotone (Dembo et al., 1991).

A.2. **Log-volume midpoint gap and its Lorentzian generalization.** *Posed by S. Sra; counterexample found by GPT-5.6 Pro (July 31, 2026); formalized by S. Sra; audited by S. Sra.*

Note: I could not locate the chat but I believe GPT-5.5 Pro already broke this earlier in 2026.

For convex bodies, $K + L = \{k + \ell : k \in K,\ \ell \in L\}$ is the Minkowski sum and Vol is Lebesgue volume. Recall that d is a *semimetric* when it is symmetric, nonnegative, vanishes only on equal

arguments, and satisfies the triangle inequality; the triangle inequality is the property that fails below. A homogeneous polynomial with nonnegative coefficients is *Lorentzian* in the sense of Brändén and Huh (2020) if all its second-order directional derivatives, taken at positive arguments, are quadratic forms of Lorentzian signature $(+,-,\dots,-)$; in two variables, a form $\sum_k a_k x^k y^{d-k}$ is strictly Lorentzian exactly when the normalized coefficients $a_k/\binom{d}{k}$ are strictly log-concave, which is the criterion used below. Lorentzian polynomials are log-concave on the positive orthant, so $F=-\log G$ is convex there and every midpoint Jensen gap J_G is nonnegative. On the cone of positive definite matrices the corresponding statement is a theorem: there $\mathsf{J}_{\det}$ is the squared S-divergence, whose square root is a metric (Sra, 2016b), and the same holds for a parameterized family of symmetric divergences on that cone (Sra, 2021).

Problem A.3 (As posed, paraphrased: S. Sra, standalone problem posed in 2012; formulation as recorded in the audit of Appendix F)**.** The log-volume midpoint gap is the square of a distance on convex bodies: that is,

$$d_{\mathrm{Vol}}(K,L) := \left[\log \mathrm{Vol}\left(\frac{K+L}{2}\right) - \tfrac{1}{2}\log \mathrm{Vol}(K) - \tfrac{1}{2}\log \mathrm{Vol}(L)\right]^{1/2}$$

is a semimetric on convex bodies.

Problem A.4 (As posed, paraphrased: S. Sra, standalone problem posed in 2019 as a Lorentzian generalization of the log-volume statement; formulation as recorded in the audit of Appendix F)**.** Let $\mathcal{C}\subset\mathbb{R}^m$ be an open convex cone, let $G:\mathcal{C}\to(0,\infty)$, and put $F=-\log G$ and

$$\mathsf{J}_G(x,y) := \frac{F(x)+F(y)}{2} - F\left(\frac{x+y}{2}\right).$$

If G is homogeneous and Lorentzian, then $d_G := \sqrt{\mathsf{J}_G}$ is a semimetric.

Both statements fall to a single witness: a strictly Lorentzian bivariate cubic, carried to convex bodies by Shephard's realization theorem. Throughout, $\mathsf{J}_G(x,y) = \log G(\frac{x+y}{2}) - \frac{1}{2}\log G(x) - \frac{1}{2}\log G(y)$ is the midpoint Jensen gap of $F=-\log G$, and $d_G=\sqrt{\mathsf{J}_G}$. The witness is isolated first, so that each of the two statements below is a reading of the same three numbers.

Lemma A.5 (The witness)**.** *Let*

$$G(x,y) = y^3 + \tfrac{11}{5}xy^2 + \tfrac{3}{2}x^2y + \tfrac{1}{10}x^3, \qquad (x,y)\in\mathbb{R}^2_{>0}, \tag{A.1}$$

and take the rational points

$$p=(35,2), \qquad q=\left(\tfrac{11}{2},\tfrac{23}{2}\right), \qquad r=\left(\tfrac{1}{500},15\right). \tag{A.2}$$

Then G is strictly Lorentzian, and

$$d_G(p,r) - d_G(p,q) - d_G(q,r) > \frac{7156704176613847159228106087505875849130467}{10^{45}} > 0. \tag{A.3}$$

Proof. Writing $G=\sum_{k=0}^3 a_k x^k y^{3-k}$ gives $(a_0,a_1,a_2,a_3)=(1,\frac{11}{5},\frac{3}{2},\frac{1}{10})$. By the bivariate criterion (Brändén and Huh, 2020, Example 2.3), strict Lorentzianity is strict log-concavity of $a_k/\binom{3}{k}$, and

$$\left(1,\tfrac{11}{15},\tfrac{1}{2},\tfrac{1}{10}\right) \text{ satisfies } \left(\tfrac{11}{15}\right)^2 = \tfrac{121}{225} > \tfrac{1}{2}, \qquad \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4} > \tfrac{11}{150}.$$

So G is strictly Lorentzian; in particular $F=-\log G$ is convex, and every Jensen gap below is nonnegative. With $R_{ab} := G((a+b)/2)^2/(G(a)G(b))$ and $\mathsf{J}_G(a,b)=\frac{1}{2}\log R_{ab}$, exact evaluation at

(A.2) gives

$$R_{pq} = \frac{609584616081}{344696430080}, \quad R_{qr} = \frac{1254036986210090216149539001}{1235460453386242764000000000}, \quad R_{pr} = \frac{614196989908809680634186 0001}{279481339600716228000000000 0}.$$

Outward-rounded rational enclosures for the three logarithms and their square roots—no floating point—then certify (A.3). □

Theorem A.6 ([**Disproof**]: log-volume distance on convex bodies)**.** *There are full-dimensional convex bodies $K, L \subset \mathbb{R}^3$ and Minkowski combinations A, B, C of them with*

$$d_{\mathrm{Vol}}(A, C) > d_{\mathrm{Vol}}(A, B) + d_{\mathrm{Vol}}(B, C).$$

Hence the proposed log-volume midpoint gap is not the square of a semimetric.

Proof. For convex bodies in $\mathbb{R}^3$ the relative Steiner polynomial is $\mathrm{Vol}(xK+yL) = \sum_{i=0}^{3} \binom{3}{i} W_i(K, L) x^{3-i} y^i$. Put

$$(W_0, W_1, W_2, W_3) = \left(\tfrac{1}{10}, \tfrac{1}{2}, \tfrac{11}{15}, 1\right),$$

which satisfies the full set of Shephard inequalities in dimension three:

$$W_1^2 = \tfrac{1}{4} > \tfrac{11}{150} = W_0 W_2, \qquad W_1 W_2 = \tfrac{11}{30} > \tfrac{1}{10} = W_0 W_3, \qquad W_2^2 = \tfrac{121}{225} > \tfrac{1}{2} = W_1 W_3.$$

By Shephard's realization theorem (1960, Theorem 4) these are the mixed volumes of some pair $K, L \subset \mathbb{R}^3$, whose Steiner polynomial is then exactly the cubic (A.1); the positive endpoint coefficients force $\mathrm{Vol}(K), \mathrm{Vol}(L) > 0$, so both bodies are full-dimensional. Setting $A = p_1 K + p_2 L$, $B = q_1 K + q_2 L$, $C = r_1 K + r_2 L$ for the points (A.2), Minkowski linearity gives $(A + B)/2 = \frac{p_1+q_1}{2} K + \frac{p_2+q_2}{2} L$ and likewise for the other pairs, so

$$d_{\mathrm{Vol}}(A, B)^2 = \mathsf{J}_G(p, q), \qquad d_{\mathrm{Vol}}(B, C)^2 = \mathsf{J}_G(q, r), \qquad d_{\mathrm{Vol}}(A, C)^2 = \mathsf{J}_G(p, r),$$

and the gap (A.3) of Lemma A.5 transfers verbatim. □

Corollary A.7 ([**Disproof**]: Lorentzian Jensen–Bregman metricity)**.** *The cubic (A.1) is a homogeneous strictly Lorentzian polynomial for which $d_G = \sqrt{\mathsf{J}_G}$ violates the triangle inequality on $\mathbb{R}^2_{>0}$. Hence $\sqrt{\mathsf{J}_G}$ is not a semimetric for every homogeneous Lorentzian G.*

Proof. Immediate from Lemma A.5: the cubic (A.1) is homogeneous of degree three and strictly Lorentzian, while (A.3) is exactly the failure of the triangle inequality for d_G at the three points (A.2). □

Remark A.8 (What the audit leaves standing). The midpoint integral identity and the homogeneous primitive decomposition of the earlier positive argument are correct. What fails is a claimed Gram-determinant identity for the derivative of a Hessian Rayleigh quotient, whose two sides have incompatible homogeneity in the line direction; the omitted calculation therefore conceals a false identity rather than a technical gap. Appendix F carries the full audit, including the exact enclosure procedure behind (A.3).

A.3. **Feasible Picard steps for DPP likelihood.** *Posed by Mariet and Sra (2015); counterexample found by GPT-5.6 (Aug 2026); formalized by S. Sra; audited by S. Sra.*

A determinantal point process with positive definite kernel L on a ground set $\mathcal{Y}$ assigns a subset $Y \subseteq \mathcal{Y}$ probability proportional to $\det(L_Y)$, the principal minor on Y. Given observed subsets $Y_1, \ldots, Y_n$, learning the kernel L means maximizing the log-likelihood

$$\phi(L) = \sum_{i=1}^{n} \log\det(L_{Y_i}) - n \log\det(I + L),$$

whose stationarity $\nabla\phi(L) = 0$ can be written as $\Delta + L^{-1} = L^{-1}$ for the residual

$$\Delta = \frac{1}{n} \sum_{i=1}^{n} U_i (U_i^* L U_i)^{-1} U_i^* - (I + L)^{-1},$$

where U_i is the indicator matrix selecting the coordinates in Y_i. The stationarity suggests the Picard iteration $L_{k+1} = L_k + aL_k\Delta_k L_k$ of the quoted passage; a step size a is *feasible* when it keeps the iterate positive definite, which the source guarantees for $a \le 1/(1-\gamma)$ with $\gamma = \max\{\lambda_{\min}(LZ), 1/\lambda_{\max}(I+L)\} \in [0,1]$. Ascent is proved for $a = 1$; the conjecture asks for every feasible $a \ge 1$.

Problem A.9 (As posed: Z. Mariet and S. Sra (2015), §2 (unnumbered))**.** Above we showed that for $a = 1$ ascent is guaranteed. Empirically, $a > 1$ often works well; Prop. A.1 presents an easily computable upper bound on feasible a. We conjecture that for all feasible values $a \ge 1$, iteration (2.5) is guaranteed to increase the log-likelihood.
The iteration carrying the step size is $L_{k+1} = L_k + aL_k\Delta_k L_k$, displayed as (2.7) in the source; feasibility is the bound $a \le 1/(1-\gamma)$ of Prop. A.1.

Theorem A.10 ([**Disproof**]: feasible-step ascent conjecture)**.** *A feasible step with $a = 5$ can strictly decrease the DPP log-likelihood.*

Proof. Use ground set $\{1,2\}$ and observations $\mathcal{T} = \{\{1\},\{2\},\{1,2\}\}$. Let

$$L_0 = \begin{pmatrix} 3.37200647 & 2.62325460 \\ 2.62325460 & 6.07953037 \end{pmatrix},$$

with the displayed decimals interpreted as exact rationals of denominator 10^8. For this data,

$$\Delta(L) = \frac{1}{3}\left[\begin{pmatrix} L_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & L_{22}^{-1} \end{pmatrix} + L^{-1}\right] - (I+L)^{-1}.$$

At $a = 5$, exact rational arithmetic gives

$$L_1 = L_0 + 5L_0\Delta(L_0)L_0 \approx \begin{pmatrix} 3.1679125391632645 & 3.1681701017272776 \\ 3.1681701017272776 & 3.4400758121589554 \end{pmatrix}.$$

Both matrices are positive definite; numerically, $\det L_0 \approx 13.6187510458$ and $\det L_1 \approx 0.8605575075$. The normalized objective is $\phi(L) = \frac{1}{3}(\log L_{11} + \log L_{22} + \log\det L) - \log\det(I+L)$. The inequality $\phi(L_1) < \phi(L_0)$ is equivalent, after exponentiating and cubing, to the exact rational comparison

$$\frac{(L_1)_{11}(L_1)_{22}\det L_1}{(L_0)_{11}(L_0)_{22}\det L_0} < \left(\frac{\det(I+L_1)}{\det(I+L_0)}\right)^3.$$

Numerically, the left and right sides are respectively

$$0.03359118941299074 \quad \text{and} \quad 0.04354945027745948.$$

Their exact rational difference is positive. Hence the step is feasible but descending. □

A.4. **Strict diagonal dominance does not ensure diminishing Nyström error reductions.** *Posed by Mark Fornace (2026); counterexample found by OpenAI Codex (2026-08); formalized by OpenAI Codex; audited by OpenAI Codex; contributed by Suvrit Sra.*

For a symmetric positive-definite matrix L and $\gamma > 0$, put $K = (L+\gamma I)^{-1}$. For a nonempty selected index set $\mathcal{I}$, the nuclear Nyström error is

$$F(\mathcal{I}) = \left\| K - K_{:,\mathcal{I}}K_{\mathcal{I},\mathcal{I}}^{-1}K_{\mathcal{I},:} \right\|_*.$$

The residual is positive semidefinite, so its nuclear norm equals its trace. The intended diminishing-error-reduction property is that, for $S \subseteq T$ and $i \notin T$,

$$F(S) - F(S\cup\{i\}) \ge F(T) - F(T\cup\{i\}).$$

This formulation avoids the convention-dependent choice between the words "submodular" and "supermodular" for a decreasing error function. The property is what would license selecting $\mathcal{I}$ greedily; without it, landmark sets are chosen by other means, for instance by determinantal sampling (Li et al., 2016) or through elementary-symmetric-polynomial design objectives (Mariet and Sra, 2017). Colbrook had already resolved Problem A.11 in July 2026, proving the SDDM case and giving exact SDD obstructions (Colbrook, 2026). The matrix below was found independently in August 2026; no priority over Colbrook is claimed.

Problem A.11 (As posed: Problem 4.6 of Amsel et al. (Amsel et al., 2026), posed there by Mark Fornace. Colbrook (Colbrook, 2026) subsequently completed the answer before this independent witness was found.)**.** Prove or disprove the submodularity of the nuclear Nyström error (14) when L is assumed, in contrast to the above, to be (a) SDDM and positive-definite or (b) SDD and positive-definite.

Problem A.11 asks whether the nuclear Nyström error retains diminishing reductions for positive-definite SDDM or SDD matrices (Amsel et al., 2026). Part (a) is true, while part (b) is false (Colbrook, 2026). Here is an independently found exact witness for part (b) with a nonempty base set.

Theorem A.12 (**[Disproof]**: SDD Nyström diminishing returns)**.** *The nuclear Nyström error need not have diminishing error reductions when L is symmetric diagonally dominant and positive definite, even when L is strictly diagonally dominant and the comparison uses a nonempty base set.*

Proof. Let

$$M = \begin{pmatrix} 11 & 3 & -4 & 3 \\ 3 & 12 & 4 & -1 \\ -4 & 4 & 14 & -2 \\ 3 & -1 & -2 & 11 \end{pmatrix}, \qquad \gamma = \frac{1}{2}, \qquad L = M - \frac{1}{2}I,$$

and set $K = (L + \gamma I)^{-1} = M^{-1}$. The diagonal entries of L minus the sums of the absolute values of the corresponding off-diagonal entries are

$$\frac{1}{2}, \quad \frac{7}{2}, \quad \frac{7}{2}, \quad \frac{9}{2}.$$

Thus L is symmetric strictly diagonally dominant with positive diagonal; Gershgorin's theorem gives $L \succ 0$.

For a selected set $\mathcal{I}$ with complement J, the block inverse identity gives

$$K_{J,J} - K_{J,\mathcal{I}} K_{\mathcal{I},\mathcal{I}}^{-1} K_{\mathcal{I},J} = (M_{J,J})^{-1}.$$

All other blocks of the Nyström residual vanish. Consequently

$$F(\mathcal{I}) = \operatorname{Tr}\big((M_{J,J})^{-1}\big).$$

Use one-based indices, take the nonempty base $S = \{2\}$, and compare adding $i = 3$ before and after $j = 4$. Direct exact inversion of the four relevant principal submatrices gives

$$F(\{2\}) = \frac{100}{349}, \qquad F(\{2,3\}) = \frac{11}{56}, \qquad F(\{2,4\}) = \frac{25}{138}, \qquad F(\{2,3,4\}) = \frac{1}{11}.$$

For example, the complement of $\{2\}$ is indexed by $\{1,3,4\}$; its 3×3 principal submatrix has determinant 1396 and the sum of its three diagonal cofactors is 400, yielding $400/1396 = 100/349$.

The two marginal error reductions are therefore

$$F(\{2\}) - F(\{2,3\}) = \frac{1761}{19544}, \qquad F(\{2,4\}) - F(\{2,3,4\}) = \frac{137}{1518}.$$

Their difference in the forbidden direction is

$$\frac{137}{1518} - \frac{1761}{19544} = \frac{2165}{14833896} > 0.$$

Hence selecting index 3 reduces the error more, not less, after index 4 has already been selected. This exactly violates diminishing error reductions. Every displayed quantity is an integer or an exact rational. □

A.5. **Statements recorded without a dossier.** These fell to the same standard of evidence as every case above — each carries a certificate in the repository that recomputes in exact arithmetic — but the statements are slight enough that the archive records them here rather than spending a section and a ledger row on each.

- A one-dimensional oblivious subspace injection with injectivity 1 can make sketch-and-solve incur the fixed factor $\sqrt{2}$ with probability 1/50. Three exact rationally weighted 2×2 sketch atoms, with exact isotropy, an exact 99/100 injection guarantee, and exact squared residual ratios. *Source:* Problem 5.1. *Certificate:* `counterexamples/osi-sketch-and-solve/`.
- The proposed NEPv matrix does not reproduce the product-state Rayleigh quotient when the Hamiltonian sum includes a same-site term with $i = j$. The certificate consists of exact 2×2 integer matrices and the integer vector $(1, 2)^T$, for which the two assertedly equal values are 0 and $-4/25$. *Source:* the identity after eq. (20) in §6.2. *Certificate:* `counterexamples/hamiltonian-nepv-identity/`.
- For an exact rational rank-three projector on eight coordinates, QRCP has strict pivots $I = (1, 2, 3)$ but the resulting inverse norm exceeds $\sqrt{18}$. An exact rational 8×3 Grassmann-chart matrix, three positive rational pivot gaps, and the Rayleigh gap 192051/250000. *Source:* Problem 4.3. *Certificate:* `counterexamples/qrcp-orthonormal-greedy/`.

Appendix B. Positive results for Sah et al's mixed-norm CS inequality

This appendix collects the positive results that bound where the mixed-norm counterexamples of Theorems 7.3 and 7.5 can live. Throughout we use the convention of Sah et al. (2020),

$$\|X\|_{p,q} := \Big(\sum\nolimits_i \Big(\sum\nolimits_j |x_{ij}|^p\Big)^{q/p}\Big)^{1/q},$$

which is the transpose of the convention used in the reported counterexamples; on the symmetric matrices $A^\mathsf{T}A$, $B^\mathsf{T}B$ and at $p = q$ the two agree, which covers every statement below. Write p' for the conjugate exponent, $1/p + 1/p' = 1$. The question is whether

$$\|A^\mathsf{T}B\|_{p,q}^2 \;\le\; \|A^\mathsf{T}A\|_{q,q}\,\|B^\mathsf{T}B\|_{p,p} \tag{B.1}$$

holds for all real $1 \le p \le q$ and all entrywise nonnegative A, B.

Lemma B.1. *Let M be Hermitian and $q \ge 1$. Then,* $\sup_{\|w\|_{q'} \le 1} w^*Mw \le \|M\|_{q,q}$.

Proof. Since $\|w\|_{q'} \le 1$ iff $\|w\|_{q'}^2 \le 1$, and $\|w\|_{q'}^2 = \|ww^*\|_{q',q'}$,

$$\sup_{\|w\|_{q'} \le 1} w^*Mw = \sup_{\|ww^*\|_{q',q'} \le 1} \operatorname{tr}(Mww^*) \le \sup_{\|W\|_{q',q'} \le 1} \operatorname{tr}(MW) = \|M\|_{q,q},$$

the last step being duality of $\|\cdot\|_{q,q}$ and $\|\cdot\|_{q',q'}$. □

Proposition B.2. *Let A, B be entrywise nonnegative and $q \ge 1$. Then* (B.1) *holds at $p = 1$.*

Proof. As $A^{\mathsf T}B \geq 0$ entrywise, $\|A^{\mathsf T}Be_i\|_1 = \mathbf{1}^{\mathsf T}A^{\mathsf T}Be_i$, so by duality

$$\|A^{\mathsf T}B\|_{1,q} = \sup_{\|u\|_{q'}\leq 1} \sum_i u_i \mathbf{1}^{\mathsf T}A^{\mathsf T}Be_i = \sup_{\|u\|_{q'}\leq 1} \langle A\mathbf{1},\ Bu\rangle \leq \sup_{\|u\|_{q'}\leq 1} \|A\mathbf{1}\|_2 \|Bu\|_2,$$

by Cauchy–Schwarz. Now $\|A\mathbf{1}\|_2 = (\mathbf{1}^{\mathsf T}A^{\mathsf T}A\mathbf{1})^{1/2} = \|A^{\mathsf T}A\|_{L_{1,1}}^{1/2}$, while

$$\sup_{\|u\|_{q'}\leq 1} \|Bu\|_2 = \sup_{\|u\|_{q'}\leq 1} (u^{\mathsf T}B^{\mathsf T}Bu)^{1/2} \leq \|B^{\mathsf T}B\|_{q,q}^{1/2}$$

by Lemma B.1. □

This result appears as Lemma 3.1 in Sah et al. (2020); the duality proof recorded there was supplied by S. Sra, and the paper's acknowledgment records as much.

Proposition B.3 (every positive integer p)**.** *If $p \in \mathbb{N}$ and $q \geq p$, then (B.1) holds for all entrywise nonnegative A, B.*

Proof. The block matrix $\left[\begin{smallmatrix} A^{\mathsf T}A & A^{\mathsf T}B \\ B^{\mathsf T}A & B^{\mathsf T}B \end{smallmatrix}\right] = [A\ B]^{\mathsf T}[A\ B]$ is positive semidefinite, so by the Schur product theorem its p-th Hadamard power is too, and is therefore $\left[\begin{smallmatrix} C^{\mathsf T}C & C^{\mathsf T}D \\ D^{\mathsf T}C & D^{\mathsf T}D \end{smallmatrix}\right]$ for some C, D with $C^{\mathsf T}D$, $C^{\mathsf T}C$, $D^{\mathsf T}D$ entrywise nonnegative. Writing $r = q/p \geq 1$,

$$\|A^{\mathsf T}B\|_{p,q} = \Big(\sum_i \Big(\sum_j (A^{\mathsf T}B)_{ij}^p\Big)^{q/p}\Big)^{1/q} = \|C^{\mathsf T}D\|_{1,r}^{1/p} \leq (\|C^{\mathsf T}C\|_{1,1}^{1/2}\|D^{\mathsf T}D\|_{r,r}^{1/2})^{1/p},$$

by Proposition B.2, and the right-hand side is $\|A^{\mathsf T}A\|_{p,p}^{1/2}\|B^{\mathsf T}B\|_{q,q}^{1/2}$. □

Proposition B.4 ($q = \infty$)**.** *For $1 \leq p$ and any complex A, B of compatible size, $\|A^*B\|_{p,\infty}^2 \leq \|A^*A\|_{p,p}\|B^*B\|_{\infty,\infty}$. Here entrywise nonnegativity is not needed.*

Proof. By duality and Cauchy–Schwarz,

$$\|A^*B\|_{p,\infty}^2 = \max_i \max_{\|z_i\|_{p'}\leq 1} |z_i^*A^*Be_i|^2 \leq \max_i \max_{\|z_i\|_{p'}\leq 1} (z_i^*A^*Az_i)\,(B^*B)_{ii},$$

and Lemma B.1 bounds the first factor by $\|A^*A\|_{p,p}$ and the second by $\|B^*B\|_{\infty,\infty}$. □

The next two statements concern the specialization $B = A$, i.e. the inequality $\|Z\|_{p,q}^2 \leq \|Z\|_{p,p}\|Z\|_{q,q}$ for completely positive Z.

Proposition B.5 (infinite divisibility)**.** *Let $Z \in \mathbb{R}_{\geq 0}^{n\times n}$ be positive semidefinite and suppose the Hadamard power $Z^{\odot p} = [z_{ij}^p]$ is positive semidefinite for some $p \geq 1$. Then $\|Z\|_{p,q}^2 \leq \|Z\|_{p,p}\|Z\|_{q,q}$ for every $q \geq p$.*

Proof. Write $Z^{\odot p} = X^{\mathsf T}X$. Then $\|Z\|_{p,q} = \|X^{\mathsf T}X\|_{1,q/p}^{1/p}$, and Proposition B.2 bounds this by $\big(\|X^{\mathsf T}X\|_{1,1}\|X^{\mathsf T}X\|_{q/p,q/p}\big)^{1/2p} = \|Z\|_{p,p}^{1/2}\|Z\|_{q,q}^{1/2}$. □

Corollary B.6 (order at most three)**.** *Let $Z \in \mathbb{R}^{n\times n}$ be positive semidefinite with $n \leq 3$, with no assumption that its entries are nonnegative. Then $\|Z\|_{p,q}^2 \leq \|Z\|_{p,p}\|Z\|_{q,q}$ for all $1 \leq p \leq q$.*

Proof. For $n \leq 3$ a PSD Z has $|Z| := [|z_{ij}|]$ PSD as well.[4] Since $p \geq 1 \geq n-2$, the FitzGerald–Horn theorem (FitzGerald and Horn, 1977) makes $|Z|^{\odot p}$ positive semidefinite, so Proposition B.5 applies to $|Z|$. The mixed norms depend only on $|z_{ij}|$, so the bound transfers to Z. □

[4]This is a classical fact about small Hermitian matrices; the source manuscript for this case attributes it to Marcus and Watkins (1969), a reference the present text has not independently checked.

Together, Propositions B.2, B.3, B.4 and Corollary B.6 confine any counterexample to non-integer p, finite q, and order at least four. Theorem 7.3 lives at $p = q = 3/2$ and order four (as a Gram matrix of two 3×2 blocks), and Theorem 7.5 at $(6/5, 6)$ and order 21 — in each case the smallest region the positive results leave open.

Appendix C. Specht modules and Borcea–Brändén's Problems 36, 37

The integer matrices of Theorems 3.7 and 3.10 are not a numerical accident, and this appendix says where they come from. They are the $t = 5$ and $t = 4$ members of one family of pencils built equivariantly on the Specht module $S^{(3,2)}$, whose $s_{(1^5)}$ coefficient is a fixed quintic in t that turns negative once t is large enough. Both problems are refuted by that single coefficient, but at different values of t, and the appendix ends by locating the two thresholds.

The module and its projections. Let M be the real permutation module with orthonormal basis $\{e_{ab} : 1 \le a < b \le 5\}$ indexed by the 2-subsets of $[5]$, and let

$$B : M \longrightarrow \mathbb{R}^5, \qquad B(e_{ab}) = e_a + e_b$$

be the $\mathfrak{S}_5$-equivariant map that forgets which pair produced which points. Row i of B has a 1 in the four columns ab with $i \in \{a, b\}$, and two rows share exactly one such column, so

$$BB^{\mathsf{T}} = 3I_5 + \mathbf{1}\mathbf{1}^{\mathsf{T}},$$

which is nonsingular. Hence B is surjective and $V := \ker B$ has dimension $10 - 5 = 5$. Young's rule for the permutation module on 2-subsets gives $M \cong S^{(5)} \oplus S^{(4,1)} \oplus S^{(3,2)}$, while the point-permutation module $\mathbb{R}^5$ is $S^{(5)} \oplus S^{(4,1)}$, so

$$V \cong S^{(3,2)}.$$

Write ρ for the representation of $\mathfrak{S}_5$ on V, orthogonal for the inner product V inherits from M. For $i \in [5]$ set

$$P_i := \frac{1}{2} \sum_{\substack{1 \le a < b \le 5 \\ a, b \ne i}} \rho((ab)), \tag{C.1}$$

half the sum of the transpositions fixing i.

Lemma C.1. *Each P_i is an orthogonal projection of rank three, $\rho(\sigma) P_i \rho(\sigma)^{-1} = P_{\sigma(i)}$ for every $\sigma \in \mathfrak{S}_5$, and $\sum_{i=1}^{5} P_i = 3I_V$.*

Proof. The transpositions in (C.1) are exactly those of the point stabilizer $H_i \cong \mathfrak{S}_4$, and their sum is the class sum of transpositions, which is central in $\mathbb{R}[\mathfrak{S}_4]$ and therefore acts on an irreducible Specht module S^{μ} by the sum of the contents of μ. By branching,

$$S^{(3,2)}{\downarrow_{\mathfrak{S}_4}} \cong S^{(3,1)} \oplus S^{(2,2)},$$

and the content sums are $0 + 1 + 2 - 1 = 2$ for $(3,1)$ and $0 + 1 - 1 + 0 = 0$ for $(2,2)$. So the operator in (C.1) is the identity on the three-dimensional $S^{(3,1)}$ summand and zero on the two-dimensional $S^{(2,2)}$ summand: an orthogonal projection of rank three. Covariance follows by conjugating the class sum, since conjugation by $\rho(\sigma)$ carries the transpositions fixing i to those fixing $\sigma(i)$. Finally $\sum_i P_i$ commutes with $\rho(\mathfrak{S}_5)$, so Schur's lemma makes it a scalar on the irreducible V; its trace is $5 \cdot 3 = 15$, hence it is $3I_V$. □

The one-parameter family. For $t \geq 0$ put

$$A_i(t) := I_V + tP_i, \qquad p_t(x) := \det\Big(\sum_{i=1}^{5} x_i A_i(t)\Big),$$

the determinant taken on V. By Lemma C.1 each $A_i(t)$ is symmetric positive definite with eigenvalues $1+t$ of multiplicity three and 1 of multiplicity two.

Proposition C.2. *For every $t \geq 0$ the polynomial p_t is symmetric, homogeneous of degree five, real stable, and has nonnegative coefficients in the monomial basis. Its Schur coefficients are*

$$\begin{aligned}
[s_{(5)}]p_t &= (t+1)^3,\\
[s_{(4,1)}]p_t &= \tfrac{1}{2}(t+1)^2(3t^2+8t+8),\\
[s_{(3,2)}]p_t &= \tfrac{1}{16}(t+1)(9t^4+48t^3+128t^2+160t+80),\\
[s_{(3,1,1)}]p_t &= \tfrac{1}{16}(t+1)(9t^4+64t^3+168t^2+192t+96),\\
[s_{(2,2,1)}]p_t &= \tfrac{1}{8}(6t^5+41t^4+108t^3+156t^2+120t+40),\\
[s_{(2,1,1,1)}]p_t &= \tfrac{1}{8}(6t^5+35t^4+96t^3+132t^2+96t+32),\\
[s_{(1^5)}]p_t &= -\tfrac{1}{16}(9t^5-21t^4-56t^3-72t^2-48t-16).
\end{aligned}$$

In particular $[s_{(1^5)}]p_5 = -743/2 < 0$.

Proof. Symmetry is the covariance of Lemma C.1: relabeling the variables by σ conjugates the pencil by $\rho(\sigma)$, which leaves its determinant alone. For real stability, suppose $z_1, \dots, z_5$ lie in the open upper half-plane and $\big(\sum_i z_i A_i(t)\big)v = 0$ for some $v \neq 0$. Then

$$0 = \operatorname{Im}\Big\langle v, \Big(\sum\nolimits_i z_i A_i(t)\Big)v\Big\rangle = \sum\nolimits_i \operatorname{Im}(z_i)\,\langle v, A_i(t)v\rangle > 0,$$

because each $A_i(t)$ is positive definite — a contradiction. Homogeneity of degree five is immediate. The coefficient of $x_1^{\alpha_1}\cdots x_5^{\alpha_5}$ is a positive multiple of the mixed discriminant of the multiset holding α_i copies of $A_i(t)$, and mixed discriminants of positive semidefinite matrices are nonnegative, so the monomial coefficients are nonnegative — here in fact strictly positive.

For the coefficients themselves, expanding the determinant exactly gives $p_t = \sum_{\mu \vdash 5} c_\mu(t) m_\mu$ with

$$\begin{aligned}
c_{(5)}(t) &= (t+1)^3,\\
c_{(4,1)}(t) &= \tfrac{1}{2}(t+1)^2(3t^2+10t+10),\\
c_{(3,2)}(t) &= \tfrac{1}{16}(t+1)(9t^4+72t^3+232t^2+320t+160),\\
c_{(3,1,1)}(t) &= \tfrac{1}{8}(t+1)(t+2)(9t^3+62t^2+120t+80),\\
c_{(2,2,1)}(t) &= \tfrac{1}{16}(39t^5+317t^4+1040t^3+1728t^2+1440t+480),\\
c_{(2,1,1,1)}(t) &= \tfrac{1}{8}(45t^5+348t^4+1100t^3+1764t^2+1440t+480),\\
c_{(1^5)}(t) &= \tfrac{1}{8}(99t^5+765t^4+2320t^3+3600t^2+2880t+960),
\end{aligned}$$

every numerator having positive coefficients, which is the promised positivity again. In the partition order $(5), (4,1), (3,2), (3,1,1), (2,2,1), (2,1,1,1), (1^5)$ the Kostka matrix of $s_\lambda = \sum_\mu K_{\lambda\mu} m_\mu$ is

$$K = \begin{pmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1\\
0 & 1 & 1 & 2 & 2 & 3 & 4\\
0 & 0 & 1 & 1 & 2 & 3 & 5\\
0 & 0 & 0 & 1 & 1 & 3 & 6\\
0 & 0 & 0 & 0 & 1 & 2 & 5\\
0 & 0 & 0 & 0 & 0 & 1 & 4\\
0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix},$$

and if c and a are the row vectors of monomial-symmetric and Schur coefficients then $c = aK$. Applying K^{-1} to the $c_\mu(t)$ gives the table above. The last line needs only the final column of K^{-1}, $(1,-2,-2,3,3,-4,1)^{\mathsf{T}}$:

$$\begin{aligned}[s_{(1^5)}]p_t &= c_{(5)} - 2c_{(4,1)} - 2c_{(3,2)} + 3c_{(3,1,1)} + 3c_{(2,2,1)} - 4c_{(2,1,1,1)} + c_{(1^5)} \\ &= -\tfrac{1}{16}(9t^5 - 21t^4 - 56t^3 - 72t^2 - 48t - 16).\end{aligned}$$

At $t = 5$ this is $-(28125 - 13125 - 7000 - 1800 - 240 - 16)/16 = -5944/16 = -743/2$. □

Remark C.3 (Two thresholds). The family crosses the two problems at different places, and the gap between them is where Theorem 3.10 lives. Schur positivity survives exactly as long as $[s_{(1^5)}]p_t \geq 0$, that is until the single positive real zero $t_2 \approx 4.2907656153$ of $9t^5 - 21t^4 - 56t^3 - 72t^2 - 48t - 16$; past t_2 the family refutes Problem 3.6. The lower endpoint of Problem 3.9 goes earlier. Since $f^{(1^5)} = 1$ and $[s_{(5)}]p_t = (t+1)^3$, its margin at $\lambda = (1^5)$ is

$$(t+1)^3 - [s_{(1^5)}]p_t = \frac{t^2(9t^3 - 21t^2 - 40t - 24)}{16},$$

so the endpoint fails exactly past the cubic's single positive zero $t_1 \approx 3.7205453558$. On (t_1, t_2) the family is therefore still Schur positive and already short of the endpoint — the substantive refutation of Problem 3.9 — and $t = 4$ is the only integer there, which is why Theorem 3.10 is stated at it. Beyond t_2 the two failures merge and stop being distinguishable, which is the reason q at $t = 5$ cannot serve as a witness for the endpoint. Both decimals are stated for orientation only: no step of the proof, and no assertion in the certificate, uses a floating-point value.

Back to the integer pencil. What remains is bookkeeping. Choose a rational basis of $V = \ker B$, assemble it as the columns of a matrix U, and let $G = U^{\mathsf{T}}U$ be its Gram matrix. In that basis P_i is no longer symmetric but G-self-adjoint, $P_i^{\mathsf{T}}G = GP_i$, so with

$$J_i(t) := 2G(I + tP_i)$$

we get $J_i(t)^{\mathsf{T}} = 2(I+tP_i)^{\mathsf{T}}G = 2G(I+tP_i) = J_i(t)$: an ordinary symmetric matrix, positive definite because $A_i(t)$ is, and at $t = 4$ and $t = 5$ an integer one in the basis the certificate picks. Since $\det(G) = 162$ there,

$$\det\Big(\sum\nolimits_i x_i J_i(t)\Big) = 2^5 \det(G)\, p_t(x) = 5184\, p_t(x).$$

At $t = 5$, writing $J_i := J_i(5)$ and dividing by 648 leaves

$$q(x) = \frac{1}{648}\det\Big(\sum\nolimits_i x_i J_i\Big) = 8\, p_5(x),$$

which is the normalization in (3.2), and $[s_{(1^5)}]q = 8\cdot(-743/2) = -2972$, the coefficient of Theorem 3.7. The factor 8 is what clears the denominators of Proposition C.2 and leaves that whole Schur expansion integral. At $t = 4$ no such factor is needed — p_4 is already integral — so $\widetilde{J}_i := J_i(4)$ and

$$\widetilde{q}(x) = \frac{1}{5184}\det\Big(\sum\nolimits_i x_i \widetilde{J}_i\Big) = p_4(x),$$

the witness of Theorem 3.10, with $[s_{(1^5)}]\widetilde{q} = 69$ against $[s_{(5)}]\widetilde{q} = (4+1)^3 = 125$. The two pencils are related by $\widetilde{J}_i = (4J_i + 2G)/5$, which is how (3.3) can be written down from the J_i and G alone.

The certificate `counterexamples/aim-problems/verify_pencil.py` recomputes all of this in exact rational arithmetic: it builds V and the P_i and checks idempotence, G-self-adjointness, rank, covariance and $\sum_i P_i = 3I$; forms both pencils and certifies positive definiteness by exact leading principal minors; expands each determinant into all 126 monomials; derives K by enumerating semistandard Young tableaux rather than quoting it, and reads f^λ off its last column rather than quoting that either; inverts to get the Schur expansions and re-extracts every coefficient independently by the bialternant formula; verifies the closed formulas of Proposition C.2 as identities

in t; and checks that $\widetilde{q}$ is Schur positive, that its lower endpoint fails at (1^5) and nowhere else, and that $t = 4$ lies strictly between the two thresholds above.

Appendix D. Additional results for McSwiggen–Sahi Theorem 2.1

Theorem 4.2 refutes Schur-convexity at $q = t$, where Macdonald polynomials are Schur polynomials and the arithmetic is smallest. Nothing about the failure needs that specialization. This appendix records the reversal for all $q, t \in (0, 1)$, identifies the lattice points at which it is admissible, locates the step at which the claim goes wrong, and says what survives.

Throughout, $n = 2$, $\lambda = (2, 0)$ and $\mu = (1, 1)$, so that $\lambda \succeq \mu$, and the evaluation point is $x = (1, 1)$. Write $t^\delta = (t, 1)$.

The rank-two ratio. Only two Macdonald polynomials are involved, and neither is quoted: in degree two the triangularity $P_\lambda = m_\lambda + (\text{lower terms})$ leaves a single unknown, and orthogonality in the (q, t) Hall inner product fixes it.

Lemma D.1. *For $n = 2$, $P_{(1,1)}(x_1, x_2; q, t) = x_1 x_2$ and*

$$P_{(2,0)}(x_1, x_2; q, t) = x_1^2 + x_2^2 + A\, x_1 x_2, \qquad A = \frac{(1+q)(1-t)}{1-qt}.$$

At $q = t$ this is $A = 1$, so $P_{(2,0)} = s_{(2)}$.

Proof. $P_{(1,1)} = e_2 = m_{(1,1)}$ is forced, being the lowest partition of 2. Write $P_{(2,0)} = m_{(2)} + A\, m_{(1,1)}$, and use coordinates in the power sums: $m_{(2)} = p_2$ and $m_{(1,1)} = e_2 = (p_1^2 - p_2)/2$. With $\langle p_\lambda, p_\mu \rangle_{q,t} = \delta_{\lambda\mu} z_\lambda \prod_i (1 - q^{\lambda_i})/(1 - t^{\lambda_i})$ we have $\langle p_2, p_2 \rangle = 2(1 - q^2)/(1 - t^2)$, $\langle p_1^2, p_1^2 \rangle = 2\big((1 - q)/(1 - t)\big)^2$ and $\langle p_2, p_1^2 \rangle = 0$, so $\langle P_{(2,0)}, P_{(1,1)} \rangle = 0$ reads

$$A\,\frac{(1-q)^2}{(1-t)^2} = (2 - A)\,\frac{1-q^2}{1-t^2}, \qquad \text{i.e.} \qquad A\big[(1-q)(1+t) + (1+q)(1-t)\big] = 2(1+q)(1-t).$$

The bracket is $2(1 - qt)$, which gives A. □

Proposition D.2. *For all $q, t \in (0, 1)$,*

$$\Omega_{(2,0)}(1, 1; q, t) = \frac{2 + A}{1 + t^2 + At}, \qquad \Omega_{(1,1)}(1, 1; q, t) = \frac{1}{t},$$

and

$$\Omega_{(1,1)}(1, 1; q, t) - \Omega_{(2,0)}(1, 1; q, t) = \frac{(1-t)^2(1-qt)}{t\,(1+t)\,(1-qt^2)} \;>\; 0. \tag{D.1}$$

Proof. By Lemma D.1, $P_{(2,0)}(1, 1) = 2 + A$ and $P_{(2,0)}(t, 1) = 1 + t^2 + At$, while $P_{(1,1)}(1, 1) = 1$ and $P_{(1,1)}(t, 1) = t$, which gives the two values. Subtracting,

$$\frac{1}{t} - \frac{2 + A}{1 + t^2 + At} = \frac{1 + t^2 + At - t(2 + A)}{t(1 + t^2 + At)} = \frac{(1 - t)^2}{t(1 + t^2 + At)},$$

the term At canceling — so the gap does not depend on A except through the denominator. Substituting A and clearing,

$$1 + t^2 + At = \frac{(1 + t)(1 - qt^2)}{1 - qt},$$

which turns the previous display into (D.1). On $q, t \in (0, 1)$ each of $(1 - t)^2$, $1 - qt$, t, $1 + t$ and $1 - qt^2$ is strictly positive. □

At $q = t = r$ this is $\frac{1}{r} - \frac{3}{1+r+r^2} = \frac{(1-r)^2}{r(1+r+r^2)}$, the computation carried in Theorem 4.2.

Where the point is admissible. The reversal above is an identity in q and t, but it refutes the claim only where $x = (1, 1)$ is a legal evaluation point, i.e. lies on $\mathcal{L}_n$.

Lemma D.3. *Let $t = q^k$ with k a positive integer and $a = 1$. Then $1^n \in \mathcal{L}_n^{q,q^k,1}$.*

Proof. Take $\mu_i = k(n - i)$, which is a partition: $\mu_i - \mu_{i+1} = k > 0$. The i-th coordinate of the corresponding lattice point is $q^{-\mu_i} t^{n-i} = q^{-k(n-i)} q^{k(n-i)} = 1$. □

So for every positive integer k the pair $\lambda = (2, 0) \succeq \mu = (1, 1)$ at $x = (1, 1)$ refutes Schur-convexity for the parameters (q, q^k), and Theorem 4.2 is the case $k = 1$. For an instance away from the Schur locus take $k = 2$, $q = 1/2$, $t = 1/4$: then $A = 9/7$ and

$$\Omega_{(2,0)}(1, 1; \tfrac{1}{2}, \tfrac{1}{4}) = \frac{368}{155} < 4 = \Omega_{(1,1)}(1, 1; \tfrac{1}{2}, \tfrac{1}{4}),$$

a gap of $252/155$.

Where the claim goes wrong. The failure is not an accident of small rank: it is the principal-specialization normalization, which does not commute with the determinant shift.

Lemma D.4 (Determinant shift for Ω)**.** *Let $\lambda = (\lambda_1, \ldots, \lambda_N)$ and $c = \lambda_N$. Then*

$$\Omega_\lambda^{(N)}(y; q, t) = \left(\frac{y_1 \cdots y_N}{t^{N(N-1)/2}} \right)^c \Omega_{\lambda - c(1^N)}^{(N)}(y; q, t).$$

Proof. From $P_\lambda(y) = (y_1 \cdots y_N)^c P_{\lambda - c(1^N)}(y)$, evaluation at t^δ contributes $\left(\prod_{i=1}^N t^{N-i}\right)^c = t^{cN(N-1)/2}$, since $\sum_{i=1}^N (N - i) = N(N - 1)/2$. Divide the two relations. □

The factor $t^{-cN(N-1)/2}$ is exactly what the argument for Schur-convexity drops. Its effect is not a perturbation. At $N = 2$ and $\lambda = (1, 1)$, where $c = 1$ and $\lambda - c(1^2) = (0, 0)$, the lemma gives $\Omega_{(1,1)}(y) = y_1 y_2 / t$, whereas the shift without the t-power would give $y_1 y_2$; the two agree only at $t = 1$. Evaluated at $y = 1^2$ this is the $1/t$ of Proposition D.2, which diverges as $t \downarrow 0$ while $\Omega_{(2,0)}(1, 1; q, t) \to 3 + q$ stays bounded. The reversal is therefore not delicate: past a point the two sides are not even the same order of magnitude.

What is not refuted. Normalizing at 1^n instead of at t^δ removes the phenomenon entirely. Write $W_\lambda(x; q, t) = P_\lambda(x; q, t)/P_\lambda(1^n; q, t)$. Since $P_\lambda(1^N) = P_{\lambda - c(1^N)}(1^N)$, the determinant shift for W carries no extra factor at all, and $W_\lambda(1^n; q, t) = 1$ for every λ — so the witness above says nothing whatever about W, both sides being 1 at the evaluation point. More than that, in the same rank-two case the W-inequality holds where the Ω-inequality fails:

$$W_{(2,0)}(x_1, x_2; q, t) - W_{(1,1)}(x_1, x_2; q, t) = \frac{x_1^2 + x_2^2 + A x_1 x_2 - (2 + A) x_1 x_2}{2 + A} = \frac{(x_1 - x_2)^2}{2 + A} \;\geq\; 0.$$

This is the rank-two shadow of a general fact on the Schur locus $q = t$, where W_λ is the normalized Schur function $s_\lambda(x)/s_\lambda(1^n)$: for every x with nonnegative entries, $\lambda \succeq \mu$ implies $s_\lambda(x)/s_\lambda(1^n) \geq s_\mu(x)/s_\mu(1^n)$, which is the conjecture of Cuttler, Greene and Skandera, proved by Sra (2016a). So on that locus the 1^n-normalized ratio is Schur-convex in exactly the sense that Ω_λ is not. Theorem 4.2 is thus a statement about the normalization, not about Macdonald polynomials on the lattice: what it refutes is Schur-convexity of $\lambda \mapsto \Omega_\lambda$, and any corresponding claim for W_λ is untouched.

The certificate `counterexamples/macdonald-schur-convexity/verify.py` recomputes all of this exactly: it recovers A from the (q, t) Hall inner product rather than quoting Lemma D.1, verifies (D.1) as a rational-function identity in q and t together with the factorization of $1 + t^2 + At$ that makes each sign visible, checks the lattice membership of Lemma D.3 symbolically in q, k and n, checks the shift exponent and the $N = 2$ instance where the naive shift fails, and confirms the W identity above.

APPENDIX E. SOME REFINEMENTS OF THE QUANTUM COUPON COLLECTION CONJECTURE

Theorem 8.4 refutes the conjecture in the Loewner order, which is what was asked. The failure is not confined to that ordering. Write

$$\Delta_n(X_1,\dots,X_n) := \operatorname{tr} Q_n(X_1,\dots,X_n) = \sum_{\emptyset\neq S\subseteq[n]} (-1)^{|S|-1} \operatorname{tr} X_S^{-1},$$

with $X_S := \sum_{i\in S} X_i$ as in the case. A positive definite matrix has positive trace, so the conjecture implies $\Delta_n > 0$, and Theorem 8.3 gives $\Delta_n > 0$ for every d and every $n \le 5$. This scalar consequence fails too, and for a reason one can see in closed form.

Let $u_1,\dots,u_n \in \mathbb{R}^3$ be such that every pair is linearly independent and every triple spans $\mathbb{R}^3$, and put

$$X_i(\varepsilon) := u_i u_i^{\mathsf{T}} + \varepsilon I_3, \qquad \varepsilon > 0,$$

so that $X_S(\varepsilon) = \sum_{i\in S} u_i u_i^{\mathsf{T}} + k\varepsilon I_3$ for $|S| = k$. A singleton leaves two regularized null directions, so $\operatorname{tr} X_S(\varepsilon)^{-1} = 2/\varepsilon + O(1)$; a pair leaves one, so $\operatorname{tr} X_S(\varepsilon)^{-1} = 1/(2\varepsilon) + O(1)$; and for $k \ge 3$ the unregularized matrix is already positive definite, so the trace of the inverse is $O(1)$. Only finitely many subsets occur, so the remainders sum uniformly and

$$\Delta_n(X_1(\varepsilon),\dots,X_n(\varepsilon)) = \frac{1}{\varepsilon}\Big(2n - \tfrac{1}{2}\binom{n}{2}\Big) + O(1) = \frac{n(9-n)}{4\varepsilon} + O(1). \tag{E.1}$$

The coefficient is negative exactly for $n \ge 10$, which the next theorem realizes with an exact witness.

Theorem E.1 ([**Disproof**]: the trace consequence)**.** *There are $d = 3$, $n = 10$ and $X_1,\dots,X_{10} \in \mathbf{S}^3_{++}$ with $\Delta_{10}(X_1,\dots,X_{10}) < 0$. A fortiori $Q_{10} \succ 0$ fails.*

Proof. Take $d = 3$, $n = 10$, $\varepsilon = 1/10000$, and, for $t = 0, 1, \dots, 9$,

$$u_t = (1, t, t^2)^{\mathsf{T}}, \qquad X_{t+1} = u_t u_t^{\mathsf{T}} + \varepsilon I_3.$$

Every X_i is strictly positive definite, and every three of the u_t are independent because their determinant is a nonzero Vandermonde product; the matrices are noncommuting, the $(1,2)$ entry of $[X_1, X_2]$ being 1. Exact rational evaluation of all $2^{10} - 1 = 1023$ nonempty-subset terms gives

$$-13901 < \Delta_{10}(X_1,\dots,X_{10}) < -13900.$$

The supplied verifier computes this with exact rational arithmetic and records the full numerator and denominator in `artifacts/certificate.json`. □

Remark E.2 (Scope). The record for the trace form is coarser than for the conjecture itself: true for $n \le 5$ by Theorem 8.3, false at $n = 10$, and undecided for $6 \le n \le 9$. Equation (E.1), whose coefficient vanishes at $n = 9$, says nothing there, since that construction is only one family; and a Q_n that is not positive semidefinite, such as the one of Theorem 8.4, can still have positive trace.

APPENDIX F. TWO NEW RESULTS FOR HYPERBOLIC POLYNOMIALS (AUTHOR: S. SRA)

These results are non-AI (by S. Sra); including here to share with interested folks.

Replacing "Lorentzian" by "hyperbolic" turns the refuted principle into a theorem. Let $p : \mathbb{R}^n \to \mathbb{R}$ be homogeneous of degree $m \ge 1$ and hyperbolic with respect to e, normalized so that $p(e) > 0$, and let Λ_{++} be the open hyperbolicity cone containing e. By Gårding (1959), Λ_{++} is an open convex cone, p is hyperbolic with respect to each of its points, and $F := p^{1/m}$ is positive, homogeneous of degree one and concave on it. The other external input is the Helton–Vinnikov theorem (Helton and Vinnikov, 2007; Lewis et al., 2005), in the normalized form: a ternary form q of degree m, hyperbolic with respect to $\mathbf{1} = (1,1,1)$ with $q(\mathbf{1}) > 0$, admits real symmetric A_1, A_2, A_3 with

$A_1 + A_2 + A_3 = I_m$ and $q(r) = q(\mathbf{1}) \det(r_1 A_1 + r_2 A_2 + r_3 A_3)$. Three points at a time is exactly what a triangle inequality needs.

Lemma F.1 (Ternary restriction)**.** *Fix $x_1, x_2, x_3 \in \Lambda_{++}$, put $T(r) := r_1 x_1 + r_2 x_2 + r_3 x_3$ and $q := p \circ T$. Then q is homogeneous of degree m and hyperbolic with respect to $\mathbf{1}$, and each coordinate vector $\varepsilon_i \in \mathbb{R}^3$ lies in the hyperbolicity cone of q containing $\mathbf{1}$.*

Proof. Homogeneity is immediate, and $w := T(\mathbf{1}) = x_1 + x_2 + x_3 \in \Lambda_{++}$ because the cone is convex. So p is hyperbolic with respect to w, and for every $r \in \mathbb{R}^3$ the univariate $\lambda \mapsto q(r + \lambda \mathbf{1}) = p(T(r) + \lambda w)$ has only real zeros; that is hyperbolicity with respect to $\mathbf{1}$. For the second claim, the path $\gamma_i(t) = (1 - t)\varepsilon_i + t\mathbf{1}$ satisfies $T(\gamma_i(t)) = x_i + t \sum_{j \neq i} x_j \in \Lambda_{++}$, so $q(\gamma_i(t)) > 0$ on $[0, 1]$: it joins ε_i to $\mathbf{1}$ inside the positive component. □

Lemma F.2 (Positive definite representatives)**.** *In that setting the matrices A_1, A_2, A_3 supplied by Helton–Vinnikov are positive definite, and for all i, j,*

$$p(x_i) = q(\mathbf{1}) \det A_i, \qquad p\Big(\frac{x_i + x_j}{2}\Big) = q(\mathbf{1}) \det\Big(\frac{A_i + A_j}{2}\Big).$$

Proof. Along the path of Lemma F.1, $L_i(t) := \sum_k \gamma_i(t)_k A_k$ satisfies $q(\gamma_i(t)) = q(\mathbf{1}) \det L_i(t) > 0$, so $L_i(t)$ is nonsingular on $[0, 1]$; the eigenvalues of a continuous path of nonsingular real symmetric matrices cannot change sign, so the inertia is constant. At $t = 1$, $L_i(1) = A_1 + A_2 + A_3 = I_m \succ 0$, hence $A_i = L_i(0) \succ 0$. The two identities are the determinantal representation evaluated at ε_i and at $(\varepsilon_i + \varepsilon_j)/2$. □

Theorem F.3 ([**Proof**]: hyperbolic Jensen–Stein pseudometric)**.** *Let p be hyperbolic as above. Then $\sqrt{\mathsf{J}_p}$, with $\mathsf{J}_p(x, y) = \log p(\frac{x+y}{2}) - \frac{1}{2} \log p(x) - \frac{1}{2} \log p(y)$, is a pseudometric on Λ_{++}.*

Proof. Concavity of $F = p^{1/m}$ gives $F(\frac{x+y}{2}) \geq \frac{F(x)+F(y)}{2} \geq \sqrt{F(x)F(y)}$, which after raising to the mth power makes $\mathsf{J}_p \geq 0$; symmetry and vanishing on the diagonal are immediate. For the triangle inequality fix $x_1, x_2, x_3 \in \Lambda_{++}$ and take $A_1, A_2, A_3 \succ 0$ from Lemma F.2. The factor $q(\mathbf{1})$ cancels, leaving

$$\mathsf{J}_p(x_i, x_j) = \log \frac{\det\big((A_i + A_j)/2\big)}{\sqrt{\det A_i \det A_j}} = \delta_{\mathrm{S}}^2(A_i, A_j),$$

the squared Stein divergence, whose square root is a metric on the positive definite cone (Sra, 2016b). Applying that triangle inequality to A_1, A_2, A_3 gives the claim, and the triple was arbitrary. □

Theorem F.4 below proves a conjecture once shared with the author (S. Sra) by Fraser (2025).

Theorem F.4 ([**Proof**])**.** *With $F = p^{1/m}$, for all a, b, c in the closed hyperbolicity cone,*

$$F(a + b)\, F(a + c) \;\geq\; F(a)\, F(a + b + c) + F(b)\, F(c).$$

Proof. Write $\mathcal{D}(M) = \det(M)^{1/m}$. We first prove the matrix form $\mathcal{D}(A+B)\mathcal{D}(A+C) \geq \mathcal{D}(A)\mathcal{D}(A+B+C) + \mathcal{D}(B)\mathcal{D}(C)$ for $A, B, C \succeq 0$. For $A \succ 0$, put $X = A^{-1/2} B A^{-1/2}$ and $Y = A^{-1/2} C A^{-1/2}$; every term carries $\mathcal{D}(A)^2$, so it suffices to show $\mathcal{D}(I + X)\mathcal{D}(I + Y) \geq \mathcal{D}(I + X + Y) + \mathcal{D}(X)\mathcal{D}(Y)$. Order the eigenvalues of X decreasingly and those of Y increasingly. Since a product does not depend on the ordering, $\mathcal{D}(I + X)\mathcal{D}(I + Y) = \big(\prod_i [(1 + x_i + y_i) + x_i y_i]\big)^{1/m}$, and Minkowski's determinant inequality $\det(R + S)^{1/m} \geq \det R^{1/m} + \det S^{1/m}$ — itself Jensen's inequality for $t \mapsto \log(1 + e^t)$ applied to the eigenvalues of $R^{-1/2} S R^{-1/2}$ — applied to the diagonal matrices $R = \operatorname{diag}(1 + x_i + y_i)$ and $S = \operatorname{diag}(x_i y_i)$ gives

$$\mathcal{D}(I + X)\mathcal{D}(I + Y) \;\geq\; \Big(\prod_i (1 + x_i + y_i)\Big)^{1/m} + \Big(\prod_i x_i y_i\Big)^{1/m}.$$

Fiedler's upper bound (Fiedler, 1971), which pairs one decreasing spectrum with the other increasing one, gives $\det(I + X + Y) \leq \prod_i (1 + x_i + y_i)$, while the second term is $\mathcal{D}(X)\mathcal{D}(Y)$; a limit $\varepsilon \downarrow 0$

in $A+\varepsilon I$ covers singular A. For the cone statement, apply Lemmas F.1 and F.2 to $a,b,c\in\Lambda_{++}$ to get $A,B,C\succ 0$ with $p(ra+sb+tc)=q(\mathbf{1})\det(rA+sB+tC)$; then $F(ra+sb+tc)=q(\mathbf{1})^{1/m}\mathcal{D}(rA+sB+tC)$ for $r,s,t\geq 0$, and the displayed inequality is the matrix one times $q(\mathbf{1})^{2/m}$. Points of the closed cone are handled by $u\mapsto u+\varepsilon e$, which lies in Λ_{++}, and continuity. □

Nothing here collides with Theorem A.6 and Corollary A.7, and it is worth saying why not. The witness (A.1) is strictly Lorentzian but *not* hyperbolic: $G(x,1)=\frac{1}{10}x^3+\frac{3}{2}x^2+\frac{11}{5}x+1$ has discriminant

$$18abcd-4b^3d+b^2c^2-4ac^3-27a^2d^2=-\frac{1499}{1250}<0\qquad\Big(a,b,c,d=\tfrac{1}{10},\tfrac{3}{2},\tfrac{11}{5},1\Big),$$

so it has one real and two complex conjugate roots. A binary form is hyperbolic with respect to some direction only if it splits into real linear factors — each factor ℓ_i contributes the root $-\ell_i(x)/\ell_i(e)$, which is real for every real x only when ℓ_i is real up to scale — so G is hyperbolic with respect to no direction at all. In two variables hyperbolicity is therefore strictly stronger than strict Lorentzianity, and the witness lives in the gap. That gap is the content of both results together: what makes the midpoint Jensen gap a squared distance is the determinantal representation that hyperbolicity buys through Helton–Vinnikov, and not log-concavity of the coefficients.